

VISUAL MANAGEMENT, PERFORMANCE
MANAGEMENT AND PERFORMANCE
IMPROVEMENT: A CASE STUDY OF
AN OIL AND GAS COMPANY

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Performance Improvement: A Case Study of an Oil and
Gas Company in Malaysia

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DECLARATION

I hereby declare that:

- (1) This Research Project is the end result of my own work and that due acknowledgement has been given in the references to all sources of information be they printed, electronic, or personal.
- (2) No portion of this research project has been submitted in support of any application for any other degree or qualification of this or any other university, or other institutes of learning.
- (3) The word count of this research report is 25026.

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LIST OF ABBREVIATIONS

BSC	Balance Scorecard
CM	Corrective Maintenance
FSO	Floating Storage and Offloading
GDP	Gross Domestic Product
GST	Good and Services Tax
KPI	Key Performance Indicator
MIDA	Malaysian Investment Development Authority
OIM	Offshore Installation Manager
OPEC	Organisation of Petroleum Exporting Countries
PM	Preventative Maintenance
RMP	Reservoir Management Plan
SCE	Safety Critical Elements
SPSS	Statistical Package for Social Sciences
SMT	Senior Management Team

ABSTRACT

As part of the downfall of crude oil commodity price, all oil and gas companies were greatly impacted. Confronted with the steep decline in oil prices and weak outlook for near-term recovery, the industry began to respond to the environment by proactively driving efficiency and making a sound decision. This study is conducted to explore how visual management and performance management can help them improve their operational performance and as a result help them to reduce their operating cost. A case study had been conducted on one of the selected integrated oil and gas companies in Malaysia. The company has been implementing performance management, visual management and scorecard since 2010. A survey was conducted and a total of 95 questionnaires were distributed to selected respondents and 91 questionnaires were returned from Operations, Subsurface, Wells and Drilling department which are the user of the scorecard and visual management. The survey measures the impact of visual management and employee perception towards the plant uptime and visibility and transparency required in managing performance by using visual management and scorecard. Data were analysed through cross tabulations and One-Sample *t* Test analysis. The results also reveal that the respondent agreed with visual management and combination of performance and visual management help to improve the performance of the organisation. An informal interview was conducted with one of the Senior Management members and four core persons from Operations department to understand What, Why and How the combination helps the company to improve their performance and the recommendations can be applied to other company in the industry. The results indicated there is a necessity to integrate performance management and visual management together in a single tool or process to achieve an optimum performance improvement results.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The oil and gas industry has been facing tremendous challenges since 2014 after they were slammed by a sharp decline in commodity price. Plunging price of oil barrels, which more than 70 percent at one point, fell, and the lowest ever hit is \$26.21 a barrel since July 2014. This had shocked the global commodity market after nearly six years of stability. The severe challenge of imbalance and volatility pressure due to the supply side had greatly impacted the oil and gas industry. As a result, it had contributed to significant investment cuts in major development projects which had a direct impact towards offsetting the natural depletion of reservoirs and ensuring a stable supply to the producers.

In this extremely challenging environment, many upstream oil and gas companies fail to survive in the market due to significant decrease in profit and increase in debt. Even for those companies that managed to survive, they had to decommission almost all drilling operations and sharply cut investment in explorations and development projects. As a result, many employees of oil and gas industry lost their jobs because companies were forced to downsize in order to reduce their operating costs.

For oil and gas companies especially in Malaysia, they are facing an additional challenge compared to others regions when our government implemented the Good & Service Tax (GST) starting 1st April 2015. This had caused most of the financial processes experience a major revamp in order to fulfil the government's new policy. As a result, possibility of poor monitoring in financial and operational performance might occur, causing the company to lose focus on what are important to the business' sustainability and growth.

Furthermore, despite the fact that the members of the Organisation of Petroleum Exporting Countries (OPEC) had come to an agreement to cut production and help firm up prices, crude oil price is still bound by the range

around US\$50s-US\$60s per barrel. This is mainly due to the U.S. shale is still continuing their production without reducing production. It's perfectly well understood that if the price of oil goes higher than the mid \$50's, then that will almost certainly bring more U.S. shale back onto the market and increase the supply problem. Thus, it will leave the oil price vulnerable to collapse back into the U\$40's rather than the high \$50's or \$60's in 2017 (Domm, P, 2016). The glorious days of US\$100 per barrel is unlikely to re-surface in the near future. In other words, oil and gas companies are still facing a challenge to maintain positive, free cash flow in this situation.

Therefore, it is very important for the oil and gas companies especially in this environment to have the processes and visibility that enable them to manage and monitor their financial health and operational performance. With this, it will allow them to survive and begin to regain their competitive advantage in this challenging climate. Moreover, the combination of performance and visual management can be one of their operational excellence tools to support the strategy of working faster, better, safer and at a lower cost.

1.2 Background of Study

In the current workplace environment, there is an increasing role of performance improvement and this has caught the attention of businesses. Why do businesses now continuously increase their focus on continuous performance improvement? This is mainly due to the ever-increasing business pressure in which the late Peter Drucker called as the era of the “*three C's—accelerated change, overwhelming complexity, and tremendous competition*”. Hence, they are required to become more effective and efficient in what they do, and do more with less in executing company strategies to remain competitive especially in the oil and gas industry. And in order to make wise tactical decisions, companies must establish a comprehensive management system that provides the visibility of critical metrics and allows them to manage their performance in order to support company's value proposition. Furthermore, organisation must continually improve

their performance by operating in a more effective and efficient way as the two are the fundamental dimensions of performance (Neely, 1995).

According to Kaplan and Norton (2006), in strategy development, strategy can be broken down as below:

“into definitive components, or goals, upon which individual employees can act. When these goals are made relevant and achievable for individual employees, they can be aligned to promote successful execution of strategy”.

Levels of transparency also require us to communicate the strategy that has been developed for the employees that impacts the organisation’s performance (Berggren and Bernshteyn, 2007). Therefore, it is crucial for organisations to create a ‘democracy of information’ to engage the workforce and allow them to act intelligently in the interest of the company when they were informed and empowered by the organisation (Hamel, 2009).

We are all aware that an effective performance management is central to the success of all organisations in implementing the organisation’s strategy, and measures how well it has met those objectives (Smith, 2014). An organisation will not be able to execute the strategy effectively and efficiently when they do not have the visibility that is required in monitoring performance where it tells the organisation what they excel in and which area they need to improve or enhance (Verbeeten, 2008). Hence, visual management contributes a significant role in assisting the organisation to simplify their reports and also to enable the visibility and transparency required in managing their performance and understanding the root cause of an issue (Bateman et al., 2016).

The intention of this research is to explore the impacts of visual management and performance management initiatives implemented in an oil and gas company since 2010 as part of the continuous performance improvement initiative. The researcher believes the initiative implemented 7 years ago, is one of the components which assists the company to remain competitive in this challenging environment of the current downturn of commodity price. Therefore, a case study will be conducted to collect evidence on the impacts and

effectiveness of the initiatives towards improving the organisation's production performance which subsequently contributes to the improvement of free cash flow. This will be done by conducting a qualitative and quantitative study on the perception of the employees from the asset and functional department after the implementation of visual management on top of performance management. The researcher believes after 7 years of implementation, this is the right time to conduct a study as there are sufficient data or evidence to perform an analysis since the results can only be seen over a period of time, not instantly. This case study will bring us the first-hand opportunity to see visual management, performance management and balance scorecard in action to realise their strengths in an oil and gas company.

Ultimately, this research is conducted to discover whether the hypotheses of performance and visual management drive performance improvement by measuring the perception of the ability of the employees using the combination of performance and visual management, in assisting them to improve the plant uptime after the implementation. A questionnaire will be circulated to the workforce at the office and worksite to find out if there is any positive relationship between these two frameworks. Adding on, an interview will be conducted with one of the senior management team members and other key personnel to find out in detail how these tools are able to help them in managing company performance.

1.3 Problem Statement

As discussed in the previous section, oil and gas industry is facing tremendous pressure in this low oil- price environment. In addition, although an effective business performance improvement initiative is one of the key requirements to sustain and to attain organisational success, the instances of failures in sustaining these continuous improvement programs are higher compared to its successes (Anand et al., 2009, De Morais et al., 2012).

Furthermore, after GST implementation by the Malaysian government, most of the organisation has become process- heavy or in other words, more

bureaucratic. As a result, organisation transparency on company strategy, performance target and actual performance is deteriorating. This has contributed to a high barrier of communication within the organisation (Berggren and Bernshteyn, 2007). Organisation performance will be further hampered when the management does not have the ability to focus on the right metric in addressing the root cause of the issue due to lack of transparency and visibility in monitoring organisation performance. The ability to monitor organisation performance can provide an early indicator before a minor issue escalates and become a major problem (Verbeeten, 2008).

Visual management techniques is widely used in lean management and operations but seldom used for strategic management purposes (Bititci et al., 2016). Despite that visual management is a simple tool, if it is properly implemented, it can help improve processes and performances dramatically. Most of the time, these visual management programs fail to provide their full potential benefit to their stakeholders because they are designed to run as stand-alone programs. Even though some of the organisations have migrated their report to balance scorecard format, most of the managers still have the perception that a scorecard is just merely a piece of report carrying no significance. Managers are holding on to their sceptical perception towards the balance scorecard when actually the tool is able to bring tremendous value to the company when properly used, allowing the manager to focus on long-term prospects of the organisation.

In addition of everything mentioned above, very little research had been conducted to find out whether it is a necessity to have a combination of both performance and visual management to achieve an ultimate result in improving the organisation performance. It appears that many researches carried a study on performance management, balance scorecard and visual management as individual entities (Youness Eaidgah, Torghabehi Alireza Arab Maki Kylie Kurczewski Amir Abdekhodae, 2016). Unfortunately, there is a limited amount of research conducted on how to establish a visual management hybrid with performance management.

1.4 Research Questions

In responding to the research problems, this research attempts to address the gap between visual and performance management. More empirical research is required to progressively bridge the gap between theoretical and practical aspects of the hybrid of performance and visual management.

The researcher aims to provide answers for the two sub-research questions identified in relation to the main research intent and topic:

1. Does visual management help to improve organisation performance?
2. Will the combination of performance management and visual management further improve organisation performance?

1.5 Research Objectives

Referring to the three sub-research questions described in the previous section, this study aims to generate evidence on how the hybrid of performance management and visual management drives performance improvement. In addition, this research will find out the responses from the management team and the perspective of the employees on the implementation of visual management.

This research attempts to achieve the following objectives:

1. To study the influence of the combination of Performance and Visual Management on a firm's performance
2. To recommend a performance improvement initiatives that can be adopted by other oil and gas company

1.6 The Significance of the Study

At present, the oil and gas industry is one of the core industries contributing to the growth of Malaysia economic. Approximately 20% of Malaysia's national GDP is contributed by the oil and gas sector, making it a mainstay of Malaysia's growth (PEMANDU, 2010) especially the national oil company Petroliam Nasional Berhad (PETRONAS). According to PWC (2016), PETRONAS had registered *“RM21 billions of after tax profit on the back of RM248 billions of revenues in 2015 compared to the RM48 billions of after tax profits from RM329 billions of revenues chalked up in the prior year”* before the plugging of oil price hit the industry.

Hence, this research will find out what other element are there that will support them to continuously improve their performance and finally re-gain their competitive advantage during this low oil price environment and, at the same time, to further enhance the development of strategic management theory. Coupled with the ability to manage and visibility of organisation performance, this can assist the oil and gas companies in Malaysia to operate in a competitive and relevant pace. In the face of these changes as well as financial pressures from rising capital costs and operational expenditures, cost control continues to be a strategic priority for most businesses.

This initiative will certainly encourage organisation to focus on their operational excellence framework that will help to improve their operations performance, improve uptime, identification of root cause and decision making. But, all these are only assumptions until a research is conducted with facts and figures to support the hypotheses. Moreover, this research is an attempt to bridge the gap, particularly in the context of an oil and gas firm setting. The outcome of this research will be able to reduce the literature gap on the knowledge of the visual management, performance management and balance scorecard and provide an adequate understanding on how these frameworks and tools able to help the organisation to improve their performance on academic focus.

Last but not least, as a Malaysian citizen and a researcher, this research will be able to contribute to Malaysia's economy especially in the oil and gas industry and more importantly to our society to witness the oil and industry to continue growing with a higher production. With that, we will see a higher national GDP as a result of higher contribution from the oil and gas industry in future.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter is a vital part of the research to review the theoretical and empirical literature related to Performance Management, Visual Management, Key Performance Indicator (KPI) and Balance Scorecard (BSC), as a foundation to better understand the issue and construct a meaningful conceptual framework for the purpose of this research. Moreover, it also serves as a critical appraisal of the most relevant and significant definitions, theories, approaches and concepts of the study.

2.2 Performance Management

Performance management is one of the topics that generate a lot of discussions due to its complexity. In the last two decades, the focus has moved from performance measurement (i.e. what to measure, how to measure and how to report the results) to performance management (i.e. how to use the measures to manage the performance of the organisation) (Bititci et al., 2011). Despite a lot of discussions pertaining to performance management, a lot of people are still confused as to what performance management is really about. Some view performance management as a typical process for planning, scheduling and budgeting. The confusion is further expanded by the fact that performance management is often recognised as business performance management by majority of people or corporate performance management and some even said it is enterprise performance management.

According to Frolick and Ariyachandra (2006), they defined performance management as *“a sequence of business processes and applications designed to improve both development and execution of the business strategy”*. By trying to integrate and accommodate various inputs and perspectives contributed by various disciplines involved, performance management is the ‘interactive closed-loop’

process where it measures the performance and used to manage and enhance organisation performance by a continuous process of optimizing the operational setting (Bititci et al., 1997). This includes assigning decision rights, prioritizing and laying out what or which goals to achieve, and measuring and assessing performance (Heinrich, 2002; Kravchuk & Schack, 1996; Brickley et al., 1995).

Ultimately, the process helps organisation in achieving optimum competitiveness and effectiveness by closing the gap between strategy and execution.

2.2.1 Performance Management Process

Performance management begins with developing a strategy derived from the mission, vision, values and the identification of strategic objectives from both individual or team and corporate level. A precise business action plan is required in meeting these objectives (Busi & Bititci, 2006; Hagos & Pal, 2010). Resources requirement needs to be considered at both levels; financial and human resources, and training and development requirement for individuals (Armstrong, 2006, 2009). Whilst on the strategy execution phase, the execution and implementation of the plan of actions and changes will take place (Eckerson, 2010).

Monitoring and Performance Measurement process is required for implementation of the action plan progress (Busi & Bititci, 2006; Parthiban & Goh, 2011). A clear goal with measurable results is mandatory to eliminate the diffusion of organisational energy (Kaplan, 2001; Rangan, 2004). By developing a set of key performance indicators (KPIs) and the ability to quantify the goals in a measurable form, this reduce the confusion of the objectives and uncertainties whether goals are achievable (Busi & Bititci, 2006). These set of measures must be balanced by having financial and operational metrics to gauge the efficiency and the effectiveness. Furthermore, a right balance is needed between measuring the actual progress (monitoring past performance/lagging KPIs) as well as measuring what used to support the look ahead (planning future performance/leading KPIs) (Bourne et al., 2003). In order to encourage

constructive behaviour from the employee, their individual target must be aligned with the organisation's strategic goals (Otley, 1999; Aguinis, 2009).

Progress for all action items from the plan should be regularly monitored to ensure the realization of the plan and objectives (Parthiban & Goh, 2011). A regular dialogue between senior management members should take place to review organisation performance and the progress of the action plan implementation. The discussion should include an improvement plan and growth action to deliver those objectives. When organisations change their strategies, they need to revise the performance measures in order to ensure they are aligned and relevant to the new company's objectives (Parthiban & Goh, 2011). In order to promote accountability at individual level, appraisal and feedback is the tool and process to reward and motivate employees towards attaining these goals and encouraging correct or positive behaviours (Otley, 1999; Armstrong, 2006; Radnor & Barnes, 2007; Barnes & Radnor, 2008; Armstrong, 2009).

Last but not least, communication plays a significant part along the entire performance management process. It is important to create commitment and alignment while promoting constructive behaviours within the organisation by applying the correct and effective communication means (i.e. how to communicate and what to communicate) (Verweire and Berghe, 2004). Thus, performance management indirectly provides a mechanism to address operational problems by communicating the strategy and in return improves the focus of the organisation to pursue their goals.

2.2.2 Performance Management Model

According to Smith & Goddard, (2002) and Franco-Santos et al., (2007), *“management researchers in this topic is from various area such as human resource management, operations management, strategy, marketing, finance, accounting, organisational behaviour, industrial economics, psychology, political science and operational research are all contributing to the field of performance management”*. As a result, a number of performance management frameworks

exist but they are industry-specific and it varies from industry to industry (Cokins, 2007). Despite the different framework from industry to industry, they still have a common structure in developing strategy, defining measures and managing performance against strategic goal by continuously fine-tuning the strategy and optimising execution of the strategy. Therefore, “performance management” is a framework that is jointly “owned” by the management research community (Thorpe & Holloway, 2008) and a conclusive “general theory” of performance management difficult to develop (Smith & Goddard, 2002).

Ultimately, all performance management models still use data analysis from the integrated data to:

1. Strategy development
2. Design robust action plan (e.g., definition of targets, models, projects, and initiatives in terms of metrics and KPIs) to implement these strategies
3. Definition, measurement, and management of performance against these metrics and KPIs
4. Adjustment of strategy and transparent communication within the organisation

Figure 2.1 illustrates the performance management model, and Table 2.1 presents a brief description of the performance management model structure.

Figure 2.1: Performance Management Model (Hossain & Prybutok, 2016)

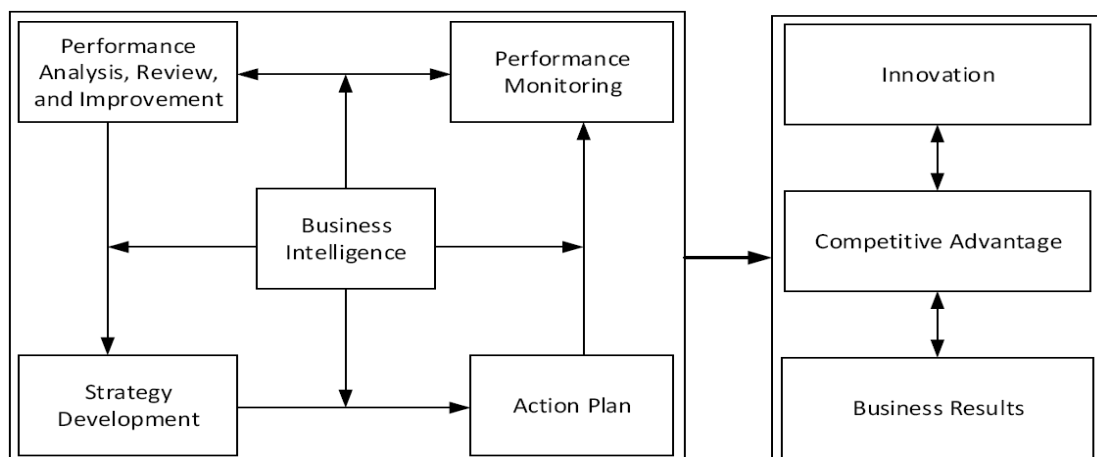


Table 2.1: Description of Performance Management Constructs

Construct	Description
Strategy development	According to Ariyachandra & Florick, (2008), strategy development is “ <i>The formulation and refinement of strategy and strategic objectives by weighing strategic challenges and advantages (i.e., strengths, weaknesses, opportunities, and threats)</i> ”
Action Plan	According to Eckerson (2004) and Ariyachandra & Florick, 2008, action plan is “ <i>The break-down of strategic objectives into discrete targets and operating models, and creation of projects and initiatives to meet these targets</i> ”
Performance Monitoring	According to Eckerson (2004) and Ariyachandra & Florick, (2008) performance monitoring is a “ <i>continuous assessing or gauging and monitoring of performance against the target or benchmark (i.e., the right metrics called the KPIs) that defined in action plan process</i> ”
Performance analysis, review and improvement	According to NIST (2003), an effective choice and use of financial and non-financial data and information is to “ <i>analyse, review, and adjust strategy and/or performance measurement with a view to achieve strategic objectives and to prepare for unexpected organisational or external changes</i> ”
Business Intelligence	According to Moss and Atre (2003), business intelligence is “ <i>A collection of integrated operational and decision-support applications and databases that provide the business community easy access to business data</i> ” (Moss & Atre, 2003) and as results support the comprehensive decision making that is intended to improve business performance (Buchanan & O’Connell, 2006).

Many companies often misunderstand that performance management is a new category to describe multiple applications running independently. These

multiple applications consist of planning, forecasting, budgeting, financial consolidating and reporting, score carding or dash boarding, business intelligence scenario modelling, and key performance indicators (KPIs) reporting. According to Eckerson (2004), he viewed Performance management as “*a common strategic and technical framework that pulls these applications together in a consistent and concentrated manner with a view to drive the whole organisation toward achievement of strategic goals and objectives*”.

2.3 Key Performance Indicators (KPI)

Key Performance Indicators (KPI) is used by organisations to monitor their processes and business performance against their business strategy (Marr et. al. 2004). The performance of the organisation is measured relative to its objective, hence allowing corrective action to be generated to address those gaps (Mate et al., 2012). KPI is usually defined in relation to the business strategy and objectives. A Balance Scorecard is used to display metrics in different angles of the organisation that need to be monitored. As a result, it provides a high-level overview of organisation status. KPI is a monitoring element that measures values against the goals and situation. Each indicator represents a target or plan numbers (a value that need to be achieved or a threshold that should not be exceeded). With these numbers, KPI will be able to show how well or poorly the measured component is performing. On the other hand, KPI will indicate the degree of satisfaction of the value compared to the target and how much it is deviating from the target (Barone et al., 2011).

Usually organisations rely on dashboards to monitor their KPI and most of the KPIs are incorporated in different dashboards which are able to provide a comprehensive view of each specific area of an organisation (Eckerson, 2010). Yet, this method causes isolation between these KPI due to zero description describing inter-relationship with each other. It provides only partial information to management as KPI was created and analysed in isolation without taking into account inter-relationships and influences between them.

In order to rectify certain problems mentioned above, researchers had proposed several solutions which are strategic and performance indicator modelling. Business Intelligence Model was recommended to support the analysis step when data repository was readily available (Jiang Lei et al., 2011). With Business Intelligence Model, it allows organisation to model their business strategy including processes in relation to every goal, indicators, and possible scenario that will impacts their goals, subsequently supporting SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis (Hill & Westbrook, 1997). By doing this, Business Intelligence Model provides a detailed understanding of the business strategy together with their relationships associated with the KPIs.

Once the business strategy had been modelled and analysed, the decision maker had a comprehensive view of the respective KPIs and the relationships between them. Thus, key performance indicators can be an effective measuring tool since it is embedded in the innermost part of the organisation performance layers (Sofiyabadia et al., 2016).

2.3.1 Lagging and leading indicators

Key Performance Indicators are also used to gauge the safety performance and usually they are differentiated by lagging and leading indicators. Number of injuries, incidents and damages are measured by lagging indicators. Whereas, leading indicators provide signals of the deviation from an ideal state by measuring safety input (CCPS, 2011). They are often represented by indicators that measure the integrity of mechanical, action items to follow-ups (i.e. corrective action from a high profile incident or safety critical repair work order) and training and competency (Reiman & Pietikainen, 2012). A management of change on a system usually happens slowly and the results can only be seen over a period of time. There a possibility on a day-to-day basis, those changes might not be noticeable. The indicators allow data to be collected over time and analysis can be conducted based on the trending identified (Vinnem, 1998).

The combinations of lagging and leading indicators are often used in the industry nowadays and both indicators complement each other. The term 'dual assurance' is used for this practice (HSE, 2006). If the outcome of the leading and lagging indicators shows deviation, this suggests the critical control mechanism does not effectively highlight the risk or perhaps the risk control system is inefficient. After all, the performance measurement indicators should be suitable and relevant. The combination indicators featuring both lagging and leading key indicators saw a gradual increase in practice to provide more effective activities tracking and safety performance monitoring. As a result, it enabled effective decision-making, encouraged corrective actions close out and helped to predict future performance (Hale, 2009). It is important to ensure the indicators are explicit, difficult to be influenced, compatible for day-to-day operations, endorsed and connected to the root cause of an incident when an organisation is developing their safety indicators (IAEA, 1999).

According to Oil and Gas Producer association, OGP (2013), the number of fatalities, fatal accident, incident rates, total recordable injury rate and lost time injury frequency have been used as lagging indicators in reporting safety performance in the oil and gas sector. Meanwhile, near misses serve as vital safety alert at offshore oil and gas installation. Despite serving as a vital safety alert at the platform, it has a limitation to display the widespread overall breakdown that indicates the present safety level of a facility even though it serves as a leading indicator. One of the historical event happened when an oil and gas industry over depended on lagging indicators and as result, a faulty process safety conditions had been overlooked and their attention had been diverted to the BP Texas City Refinery explosion (Baker, 2007). After that incident, the industry began to focus on proactive safety monitoring. They have now started to measure the efforts and preventive measuring that commit to safety management and accident prevention. Leading indicators have been implemented dramatically by the oil and gas industry on top of the lagging indicators. This is to examine the reliability of the safety barriers or to understand the requirement of strengthening those protection barriers and remedial actions planning for safety systems (Reiman & Pietikainen, 2012).

A good indicator system would usually adopt multiple approaches by using both lagging and leading indicators that include process safety, potential catastrophic accidents, personal risk and preventative measures (Tang et al., 2017). The need to enhance process safety indicators is emphasised by the safety review panel and would remain unchanged across the industry to prevent any catastrophic accidents (Baker et al., 2007). By implementing common indicators, this will enable the platform to be a benchmark in the safety performance with other platforms and subsequently enable improvements to happen (Ettorchi-Tardy et al., 2012).

2.3.2 KPIs on Scorecards and Dashboards

Scorecard and dashboard are often used as a mechanism to display or convey performance information at one glance to the reader (Baker, 2015). A dashboard or scorecard feature almost entirely is KPI's which is linked to corporate strategy that is tailored to each individual and organisation based on their role and level (Eckerson, 2010). The KPI's data displayed on the scorecards or dashboards allows organisation to monitor their performance and subsequently permit improvement to be made by providing information that support decision making.

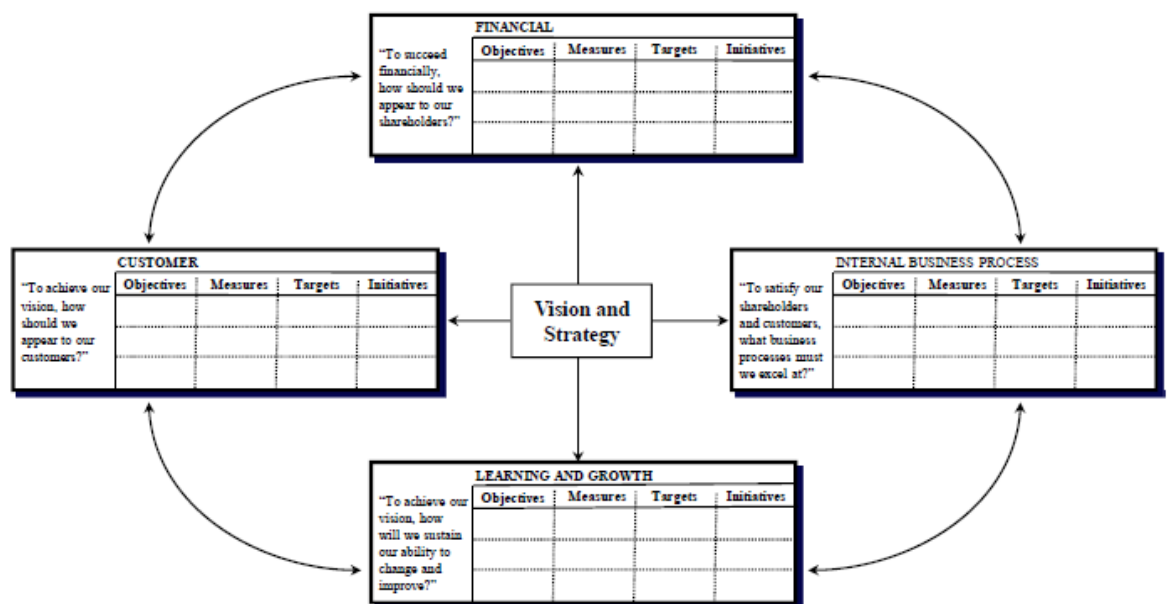
Granularity of the KPI data displayed on the dashboard or scorecard must be able to provide adequate visibility and transparency. This is required to support the drill-down capabilities for knowledge acquisition through process analysis or data mining. In return, it will be able to support the analysis, evaluation, and possible actions associated with the potential decisions (Karami et al., 2013).

2.4 Balance Scorecard (BSC)

Balance scorecard was introduced by David Norton and Robert Kaplan on their Harvard Business Review in 1992. It was developed after a year's research activity in 12 companies through a performance measurement framework (Kaplan

& Norton, 1992). The problem with traditional performance measurement that most researchers are concerned about is that they only focus on financial metrics (Johnson & Kaplan, 1987). Thus, such measures will restrict managers to only focus on short-term financial results, making the long-term prospects of the organisation neglected. Subsequently, a performance pyramid is introduced with the objective of “balance” to combine the measures of financial and non-financial performance (McNair et. al., 1990). In 1992, scorecard was introduced with the idea of “balance” and the solution was to address the concern on traditional performance measures. The conceptual foundation of the balance scorecard was developed based on performance measurement studies where the organisation’s intangible assets played a crucial role in value creation (Nolan, 1991). Figure 2.2 explains the original architecture of the balance scorecard. The balance scorecard contains financial metrics as the critical measurement for company’s success and supplement by three additional angles – customer, internal process, and learning and growth which proposed to drive the creation of long-term shareholder value.

Figure 2.2: Original Structure of Balance Scorecard – Four Perspectives



David and Robert's interest in measurement for driving performance improvement was triggered by a prominent British scientist, Lord Kelvin, centuries ago.

I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind. If you cannot measure it, you cannot improve it. (Kelvin, 1883).

2.4.1 Balance Scorecard: Translating Strategy into Action

Kaplan and Norton in their book “The Balance Scorecard: Translating Strategy into Action” had re-classified “internal business perspective to internal business process with addition of innovation element and the innovation and learning perspective to Learning and Growth” in 1996. The balance scorecard's 1996 version was different from 1992 in the sense that now the scorecard contained the result of the measures and the outcomes of the performance drivers which linked with cause-and-effect relationship, creating the performance measurement system a feed-forward control system. First, the intangible assets value was indirect. For example, knowledge and technology rarely have a direct effect on revenue. But it will impact the financial results via chain of cause-and-effect relationships involving two or three intermediate stages when there is an improvement. Below are the examples of service management linkages in profit chain that best describe the cause-and-effect relationships in these intangible assets.

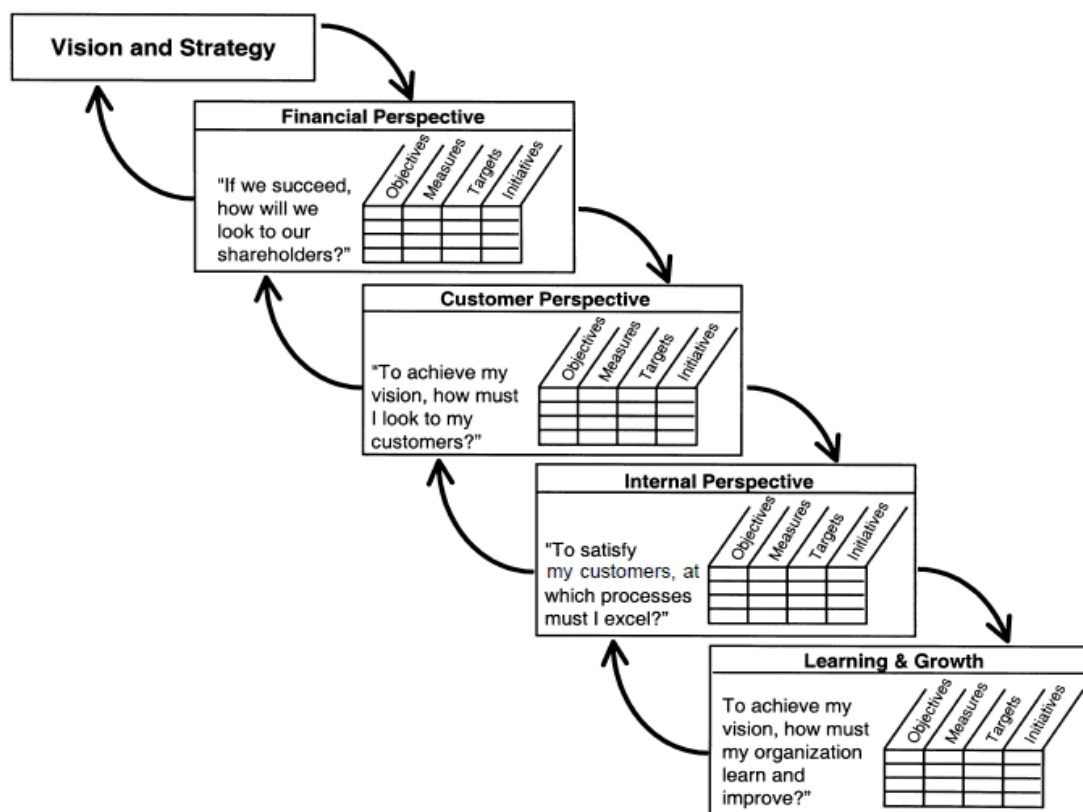
- investments in employee training improves the service quality
- better service quality results to higher customer satisfaction
- higher customer satisfaction leads to increased customer loyalty
- increased customer loyalty generates increased revenues and margins

The intangible assets rarely provide value by themselves and to generate value they usually need to combine with other intangible or tangible assets

(Kaplan, 2009). Moreover, it relies heavily on the organisation strategy where it is implemented to generate value. They believe, integrating the measurement of intangible assets into organisation management system will help organisation in improving their management on their intangible asset.

The article released in 2001 by Robert S. Kaplan and David P. Norton focuses on how strategies can be clearly connected to the standpoints by way of mapping. Different organisation's balance scorecard will be embedded into a cause-and-effect chain on a strategy that links to the expected outcome with the outcome drivers (Kaplan & Norton, 2001). The aim here was to outline personal and group objectives and create a component within a comprehensive management system that integrates strategy and operations, thus, it created a strategic awareness. At the end of the day, this will help to align the organisation and to create synergies.

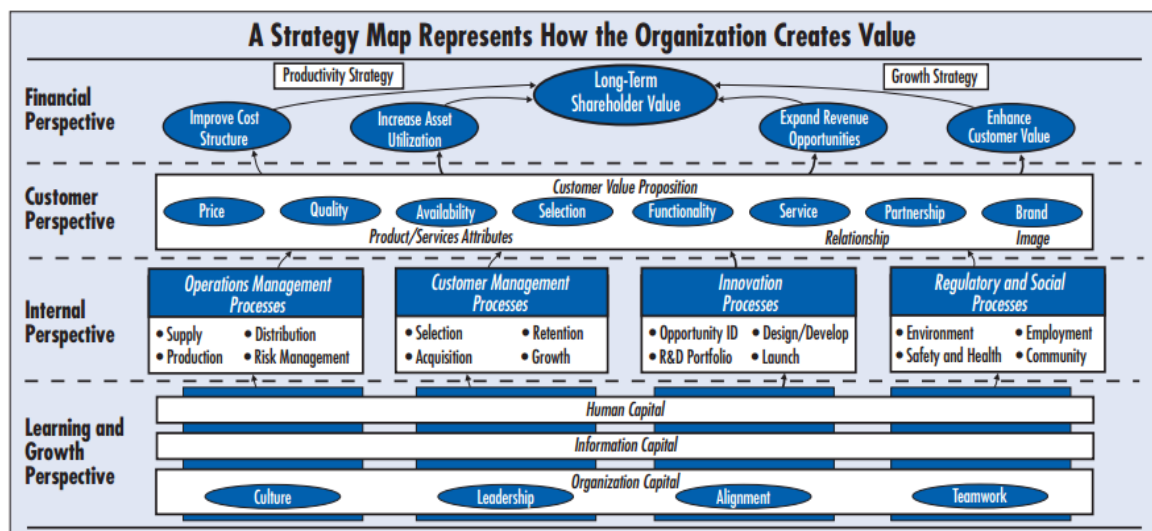
Figure 2.3: Balanced Scorecard Defines a Strategy's Cause-and-Effect Relationships



2.4.2 Balance Scorecard: Strategy Map

A new framework “Strategy Map” was invented by Robert S. Kaplan and David P. Norton in 2004 from their continuous research. This framework is a logical and comprehensive structure to explain organisation strategy as explained in Figure 2.4. The strategy map lets organisations define the connection between intangible assets and value creation. Furthermore, it permits organisations to specify the critical elements in connection to the organisation’s strategy (Kaplan & Norton, 2004). As a result, managers will be able to align their investments in people, technology and organisation capital for a greater outcome. Moreover, with this tool, everyone in the company can understand what works and what does not.

Figure 2.4: Balanced Scorecard Strategy Map



In order to create a mutual understanding and a single point of reference across the entire organisation, converting their strategy into logical architecture of a strategy map and Balanced Scorecard is required.

The next evolution of scorecard by Robert S. Kaplan and David P. Norton was in 2006 where they noticed that business units must be aligned if they want to save time and money. According to Kaplan & Norton (2006), to reduce waste and increase opportunities, organisation needed to ensure all business units were fully

aligned. When organisations have conflicting goals, that is where companies will experience difficulty in realizing their potential and improving their performance.

The idea of alignment in a balanced scorecard or a strategy map framework was mainly driven by “Management by Objectives” from Peter Drucker, the most famous management researcher. Drucker introduced management by objectives in his book “The Practice of Management” 1954 and claimed that all employees should have personal performance objectives that aligned strongly to the company strategy:

“Each manager, from the “big boss” down to the production foreman or the chief clerk, needs clearly spelled-out objectives. These objectives should lay out what performance the man’s [sic] own managerial unit is supposed to produce. They should lay out what contribution he and his unit are expected to make to help other units obtain their objectives. [...] These objectives should always derive from the goals of the business enterprise. [...] [M]anagers must understand that business results depend on a balance of efforts and results in a number of areas. [...] Every manager should responsibly participate in the development of the objectives of the higher unit of which his is a part. [...] He must know and understand the ultimate business goals, what is expected of him and why, what he will be measured against and how” (Drucker, 1954)

Robert and David believed that the organisation higher management should be responsible and accountable for organisation alignment. By using the step-by-step approach of the Balanced Scorecard management system, this will assist them to coordinate the implementation of corporate strategy (Kaplan & Norton, 2006).

Balance Scorecard has been widely received at Harvard Business Review since its origin in 1992. It has been branded as one of the “most effective management ideas of the 20th century”. Nevertheless, the balanced scorecard has evolved to keep up with the changes in the robust nature and interest of business.

2.5 Visual Management

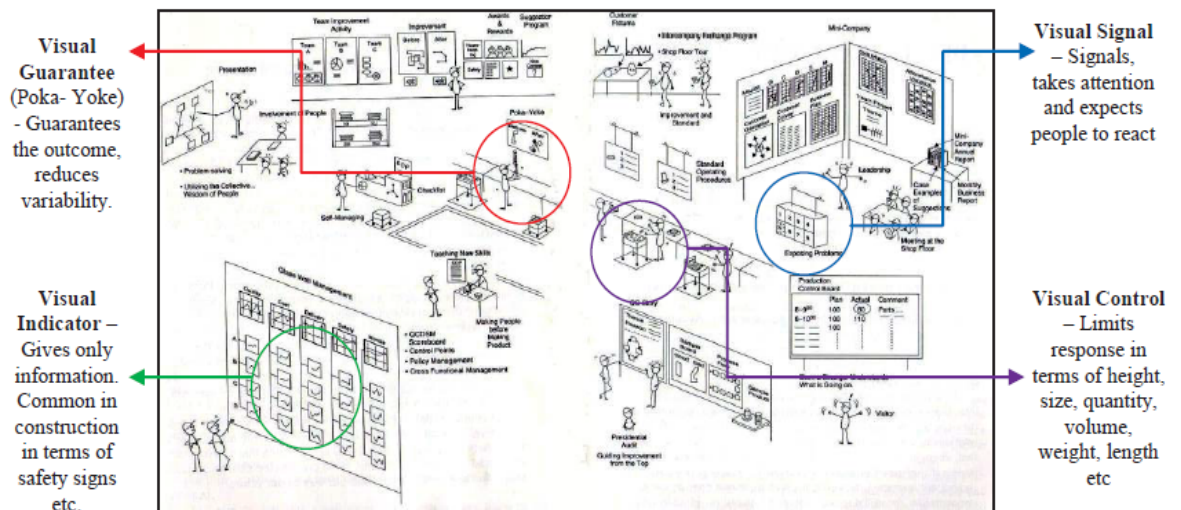
The application of visual methods in this new era management was mainly embedded in the Toyota production system and Lean Management theory (Murata & Katayama 2010, 2013; Tezel et al., 2009). In Lean Production Systems, Visual Management is one of the fundamental components which provided the basis for other improvement methods. Visual management system was designed to display problems or abnormalities in a process immediately apparent. Any new employee entering the work place, would quickly see what is going on with the capabilities to easily interpret it and see what is under control or deviates, including those who are unfamiliar with details of the processes. When there is an abnormality, the deviation is visible to all so that corrective actions can be taken immediately to the way things should be. Visual management also was a method to ensure visibility of the processes to everyone at the workplace and supports the flow of work. It used a simple tool to show the machine status, the plan of the entire operating area or definable objective and the resources (Galsworth, 1997). Visual tools can be in many forms such as white boards, charts, images, process maps, checklists and even sticky notes.

The belief behind visual management was based on the assumption that the workforce are attracted to something by the experience they see (Ho, 1999). Studies have shown that humans are better at handling visual information than text. The outcome of one of the research conducted at 3M Corporation showed that humans can digest visual data 60,000 times faster than text (Harley, 2001). Further research discovered that the human brain is capable to decode images simultaneously (in parallel). However, to decode and process language, it can only be done in a linear and sequential manner and therefore, it required more time to process language versus image (Cohen et al, 2002).

A properly- designed visual management system need to include an action component that is connected to the visual tools where an instant feedback will be provided to workers. According to Galsworth (1997), Visual Management can be successful in a virtual workplace that is designed “*to distribute information, signaling, limiting or guaranteeing (mistake-proofing/ poka-yoke) visual devices*

to communicate with workers, so that places become self-explanatory, self-ordering, self-regulating and self-improving”. An example of a visual workplace practicing visual management is shown in Figure 2.5.

Figure 2.5: Visual Workplace (Adapted from Suzaki, 1993:14-5)



2.5.1 Origins of Visual Management Systems

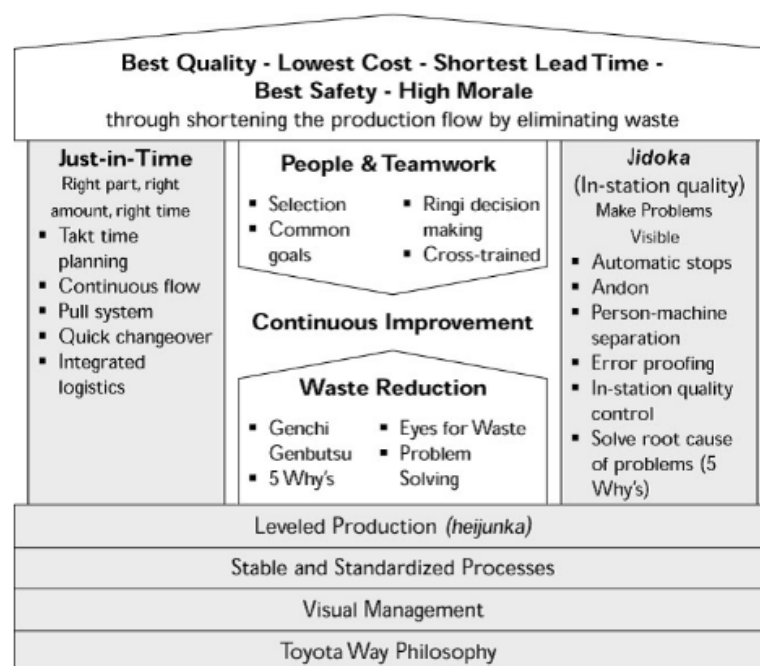
Make your workplace into a showcase that can be understood by everyone at a glance. In terms of quality, it means to make the defects immediately apparent. In terms of quantity, it means that progress or delay, measured against the plan, and is made immediately apparent. When this is done, problems can be discovered immediately, and everyone can initiate improvement plans. ~Taiichi Ohno, founder of Toyota Production System

Visual management value has been apparent since the introduction of Toyota Production System – Toyota’s revolutionary “socio-technical system for manufacturing” developed since 1948 until 1975.

It all started around 1935, when the founder of Toyota Motor Corporation, Kiichiro Toyoda, authored a 10-centimeter-thick manual on work flow processes and distributed to key employees (Hino, 2006). Taiichi Ohno, was a machine shop

supervisor who answered to Kiichiro's vision by creating new methods for manufacturing called Toyota Production System (TPS). It was described by Ohno (2003), *“the most important objective of Toyota Production System (TPS) is to increase production efficiency through consistency and by eliminating waste”*. According to Liker (2004), Toyota Production System (TPS) also broadly *“hybrids Visual Management in its operational and managerial activities”*. The framework for Toyota Production System is illustrated in Figure 2.6 below.

Figure 2.6: The Toyota Production System Model (Adapted from Liker, 2004)



2.5.2 Visual Management Functions

A detailed review on the five components of Visual Management functions are outlined briefly by Tezel et al. (2009) as follows:

2.5.2.1 Transparency

Ambiguity in work processes and underperforming staffs can cause frustrations in organisations, which potentially can lead to conflicts, staffs

engaging in disruptive behaviours and creating a negative working environment. As result, the organisation often requires added process flows (like quality checks) or added tasks (like multiple sign-offs) due to absence of trust on the work processes or the people within that process. One of the root causes to these problems can often be found in the lack of transparency.

Visual management supports transparency in the work place and relays valuable information to the workforce in an organisation. According to Formoso et al., (2002), transparency in Visual Management is regarded as the capability of the work processes to communicate with people. According to Koskela (1992), transparency can be attained when the main process flows visibly and is understandable since the beginning until the end through public display of information, measurement, and physical means. Visual management also helps in making a problem visible, allowing fast response to resolve the problem. As a result, it motivates the workers to seek for improvement by providing them the visibility of errors.

2.5.2.2 Continuous improvement

According to Bessant and Francis (1999), Kaizen was the effort taken by employees at every levels in an organisation that focused in seeking constant incremental innovation over time. Any part of ongoing effort to improve products, services, processes or productivity is a “Kaizen” or Continuous Improvement process. Visual management was a core element that helped sustain endless improvement (Suzaki, 1993; Imai, 1997). It helps to stimulates employees’ participation in all processes for management and quality improvement (Greif, 1991; Schonberger, 1992). (Greif, 1991; Schonberger, 1992). An effective visual management system encouraged employees at all levels to think about ways to continuously improve and allow the organisation to perform at its highest levels.

By using the right tools, visual management was able to support continuous improvement by displaying ideas on the idea board (Mann, 2005). This assisted employees by providing them the techniques to resolve the problem

(e.g. the seven basic tools for problem solving) (Imai, 1997), that summarised the problem solving process (e.g. A3 sheets) (Shook, 2008), and acknowledged the involvement in continuous improvement (e.g. superstar boards) (Liff & Posey, 2004).

2.5.2.3 Job facilitation

To relieve people's effort on their daily tasks relevant visual aids through job facilitation was provided. Presenting an instance with complete and precise understanding of job requirement that can assist people in executing their routines, this is what we call visual management (Greif, 1991; Suzaki, 1993; Galsworth, 1997). Galsworth, (2005); Ortiz and Park, (2010); Kattman et al., (2012) it was found that searching, counting, answering, asking, testing is an unnecessary human intervention and can be reduced by integrating visual aids in the workplace since it does not provide any value to the end results.

Label, shape, texture, size, location, label and colour coding are commonly used by people most of the time as a technique for job facilitation in visual management (Sanders and McCormick, 1993; Helander, 2006; Lehto and Landry, 2012).

2.5.2.4. Management by Facts

According to Gunasekaran et al. (1998), Management-by-facts asserted the usage of actual data in numerical form or statistics. One of the features of Visual Management is to share the organisation's actual performance or reality through information movement by displaying the information to the pertaining groups of people. Based on the research by Grief, (1991) and Galsworth, (2005) this helped to remove the monopolies of information at certain level and force employees to confess their own performance reality. There would be no personal bias in the shared information and furthermore, it is uninhibited by any subjective experience. As a result, it created a sense of openness and common objectives

amongst employees to improve performance and quickly deliver the results (Clark and Payne, 1997; Lewicki et al., 1998).

2.5.2.5 Simplification

To manage information in this dynamic and complex business environment sometime can be beyond the individual abilities. Strategic information was mainly used by organisation to interpret changes in their exterior surroundings or decision making (Choo, 1996). Organisations will also use strategic information to generate new information and knowledge from organisation learning. Mismanaged information may result in conflicts, disagreements, poor performance, stress, waste and other inefficiencies. Visual management is a mechanism to continuously monitor, filter, simplify and distribute vital information to individual employees and teams within an organisation. (Suzaki, 1993; Liff & Posey, 2004; Galsworth, 2005). The visually displayed information is structured and condensed and it represents the organisation's strategy and business objectives for employees to use as a guide in their daily routines.

2.5.3 Application of Visual Management

Many service industries have adopted the methodology with the aim of improving their processes and increasing customer satisfaction. Some of the visual management tools are widely available for implementation and they are highly standardised across the universe. Moreover, these visual tools can be tweaked by the organisation to suit their requirement of working environment.

A study done by Valente et al. (2016) concerning different areas of expertise had concluded that, the majority of organisations implementing visual management were within the areas of construction, manufacturing, project management and business. However, visual management is currently practiced and applied by various knowledge fields such as social sciences, neuroscience and

visual languages and computing. Thus, it is proven that visual management is more than making charts and metrics visible. It is about standardizing work processes and developing the workforce with critical thinking ability so that they can solve those problems in real time and create efficiency.

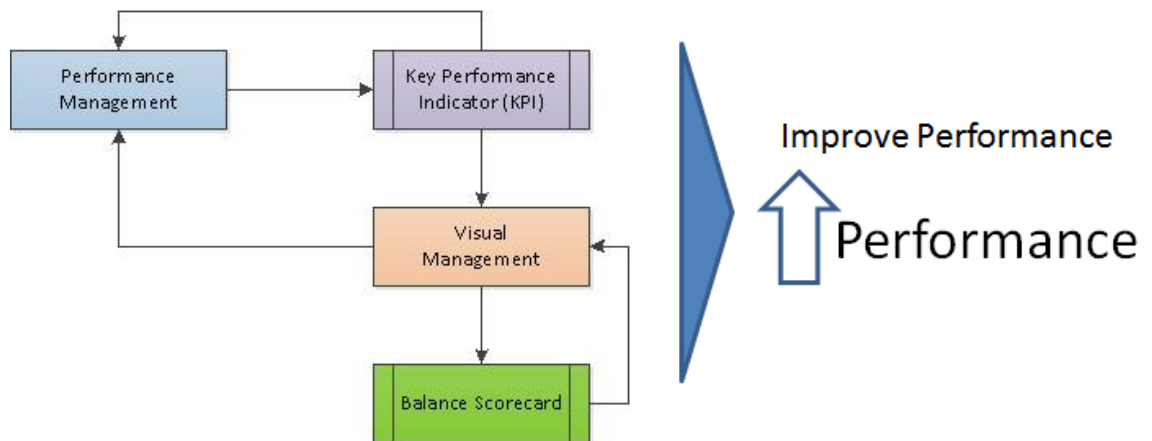
In business communication, Visual Management is often used to explain, simplify, highlight or recap certain issue/topic, reinforce, unify, and attract or make people excited (Bovee & Thill, 2005). One of the challenges in distributing the information to multiple departments is to establish the common understanding. Visual Management was used to increase the awareness of the working environment in different departmental units and organization atmospheres (Greif, 1991; Suzuki, 1993). It is able to remove the unfavourable effects of fragmentation and organisation boundaries by acting as a unifier. It helps increase sensitivities of their people towards the environments by creating a “borderless” organisation where people can act and communicate transparently (Rastogi, 2000).

2.6 Theoretical Framework

A theoretical framework is a collection of interrelated concepts and it is critical in exploratory studies. The framework will serve as a guide for the researcher throughout his entire research journey to ensure the researcher measures the correct variable and look for a correct statistical relationship. .

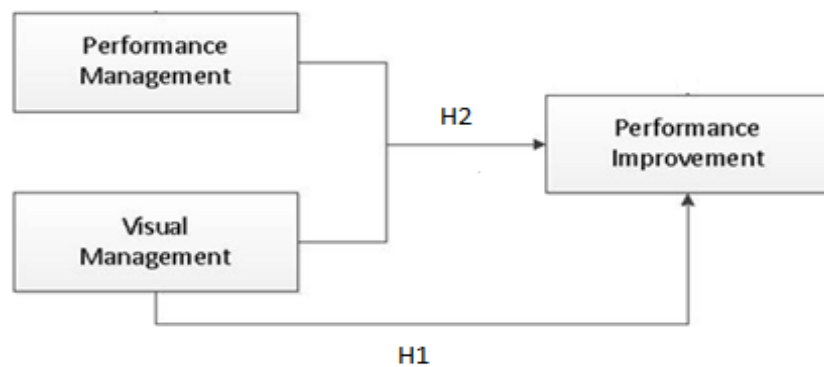
The research framework (Figure 2.8) structured below is the outcome from the literature review of Performance Management, Visual Management, Key Performance Indicators (KPI), Balance Scorecard and the details of study analytic models relevant to the same industry. As a result, the “Cases” and “Variables” are based on appropriateness, ease of application, and explanatory power of this research.

Figure 2.7: Theoretical Framework



2.7 Research Framework and Hypothesis of the Study

Figure 2.8: Research Framework



Hypotheses 1: Visual Management will improve the operational performance.

Hypotheses 2: Performance Management hybrid with Visual Management will further improve performance.

CHAPTER THREE: RESEARCH METHOD

3.1 Introduction

This chapter describes the methodology and design adopted for this research. The research design will act as guidebook for the researcher to identify the most suitable research methodology and the methods that should be used to collect and analyse the data (Collis & Hussey, 2013). The rationale for the selected methods used within this research also includes the measures taken to safeguard the ethicality and reliability of the research.

This study is essentially an in-company research. The setting for the study is a multinational integrated upstream and downstream oil and gas firm that has become increasingly volatile, uncertain and competitive nowadays. The energy sector, particularly in the domain of the exploration and production of oil and gas reserves, operates under a relatively volatile business environment since the law of demand and supply is driving the commodity price including geopolitical forces.

The study examines the employee perception and effectiveness of past implementation of performance and visual management on the selected organisation to gauge the impact towards their performance. For the intent of this research, a mixed method methodology with the combination of quantitative and qualitative methods was selected as a means to reinforce and substantiate findings from the study. The data that collected to perform the analysis are based on variables identified from the balance scorecard. The data is subsequently analysed using SPSS for Crosstabs and One-Sample *t* Test. On the qualitative part, a questionnaire will be developed and an interview will be conducted based on the questionnaire. This is to provide a perspective of the employees who utilize the tools that will be implemented.

3.2 Research Design

A research design is a structure or layout of a business research that is defined as exploratory, descriptive, or the causal research (Zikmund, 2003). Cooper & Schindler, (2003) defined research design as an overall plan that provides overall view of the research. In this plan, it contains the summary or structure of the research that the researcher will utilize to conduct the hypothesis writing and operational consequences to the final analysis of the data. It also serves as procedure collecting and analysing the required information to perform the research and act as a guideline to improve the value and reliability of the research.

In this study, the researcher will use mixed method to conduct this research. Mixed method research uses the combination of qualitative and quantitative approach. The process involves collecting, analysing and integrating quantitative (survey) and qualitative (interviews) research. This approach aims to provide a detailed understanding of the research problem. Recognize that all research methods have their own limitations; hence this can be overcome by using mixed method to offset the flaws inherent in applying each approach by itself (Creswell & Piano, 2011). Furthermore, the research problems can become research questions that will be able to address theoretical perspective at different levels. As a result, this will further strengthen the reliability of the research.

Table 3.1: Comparison between Qualitative and Quantitative Methods

	Quantitative	Qualitative
Strength and weakness	Survey	Interviews
	Generalization (Weakness)	Details, in depth (Strength)
	Large sample (Strength)	Small sample (Weakness)

The researcher uses mixed methods to conduct this research mainly due to one data resource is not enough to understand the perception of the employees on how performance and visual management will be able to drive performance improvement or otherwise. The initial findings of the quantitative result need to be further explained and justified by qualitative method on how the framework is able to bring continuous improvement culture into the organisation. The interview conducted will be able to provide an understanding on employee perception towards the process and tools that had been implemented. By doing that, it helps to confirm or support the outcome acquired from different methods. Subsequently, the qualitative result will justify why and how performance and visual management will impact performance in the organisation.

As a result, this research method will provide a better understanding of the research question from other points of view and explain unanticipated outcomes and/or possible conflicts.

3.2.1 Inferential Statistics

In descriptive statistics, researchers attempt to describe the numerical characteristics of their data. For inferential statistics, researchers attempt to go beyond their data to generalise the findings to the larger population (Johnson & Christensen, (2017). Inferential statistics will be use by the researcher to forecast or predict (inferences) from the data collected based on the characteristics of the sample from the probability of characteristics of the population. The strength of the correlation between the independent (causal) variables, and the dependent (effect) variables will also be evaluated.

Moreover, this method is used to fulfil the requirement of a mixed model that the researcher need to conduct prior and after the implementation to determine whether the performance management hybrid with visual management works to improve performance. With this, the researcher will be able to understand the probabilities of success prior to making any recommendations which is worth the time and efforts of the researcher.

One of the areas of inferential statistics is to estimate the parameters by taking the statistic from the sample data. The data will be used to describe the population parameter (i.e. the population mean). The second area is to conduct a hypothesis testing. This is to test specific hypotheses about the population based on their sample data and to answer the research question. To outline the findings of the sample including standard deviation, mean, or boxplot and explaining the characters of the probability distribution, the researcher will use descriptive statistics.

3.2.2 Qualitative Method

The qualitative study objective is to understand a certain topic in detail which includes the underlying reasons why the outcome is in such a way. In general, the qualitative research looks for answers as to why, when, what, when and how (Crowther & Lancaster, 2009). Some of the researchers agreed that the qualitative research can overcome weaknesses of the quantitative research (Cooper & Schindler, 2006)

Qualitative data collection methods can be unstructured or semi-structured techniques. This can be conducted through methods like interviews or focus group discussions and in a typically small sample size or less respondents in fulfilling the target sample size. Purposive sampling will be used to collect the data from this interview. For this research, the target sample will be the employees involved in implementing and operating/using the visual management.

In this research, this method is used to see an answer to a question on what, how, and why performance and visual management will impact performance in the context of behaviours and motivations. Questions will also be asked about the views of employees on the process of performance management hybrid with visual management. Those answers will act as evidences to support the finding on the quantitative side.

3.2.3 Quantitative Method

According to Neuman (2007), quantitative study is a comprehensive empirical exploration of social phenomena over statistical and calculated techniques. Quantitative study is normally assumed as a scientific method in social science research (Tewksbury, 2009). It was used to measure the problem by producing mathematical data that can be converted to statistics. Once the data are converted to statistics, they can be used to clarify certain phenomenon and generalize across a group of individuals. When no numbers are involved, then it is not quantitative research.

The goal of quantitative study is to develop the models, hypotheses and theories relevant to the social phenomena. Moreover, the quantitative research is used to identify precise measurement for certain things, and normally the survey forms will pose questions asking how many, how much and how often (Cooper & Schindler, 2006). Measurable data will also be used to formulate facts to uncover certain patterns in the research. The data collection method is structured and can be conducted through surveys or close-ended information such as rating scales or performance instruments. To analyse this type of data, scores collected from instruments (survey questions) or checklists will statistically be evaluated to answer the investigation queries or to confirm the assumptions.

For this research, a questionnaire will be designed and divided into two parts. The first part is to evaluate the target sample perceptions and their general understanding on performance and visual management before moving on to a more specific question. In the second part, the questionnaire will be designed to focus on a specific area and tools that had been implemented on the selected organisation. The objective is to analyse and understand the impact and effectiveness of the combination of performance and visual management in improving operational performance. With this, it will allow us to witness the responses from the users themselves whether do we need to reject the hypothesis that had been outline by the researcher.

3.2.4 Scope of Study and Population

Since this is an in-company research, hence, the population of this research will be one of the selected integrated Oil and Gas Company. The population selected for this study is the group of permanent and direct-hire employees working onshore and offshore as the direct users of the performance visual management tool in the selected organisation. Samples will also be drawn from selected departments with direct involvement on the process of prior and after the implementation of performance and visual management on the selected organisation. Furthermore, the selected departments are the core contributors for the core business performance which is Crude and Gas production. The selected departments are Operations, Subsurface, Drilling and Wells.

3.3 Data Collection Method

In this research, both forms of data will be collected by the researcher at the same time. Both data will be merged in the analysis of the overall results to provide concrete summary and evidence. The data collection will involve gathering both numeric information (e.g. survey) and qualitative response (e.g. interviews) based on the designed questionnaire. Each method has been verified to ensure the right method is selected for this research and compatible since mixed method is used for this research.

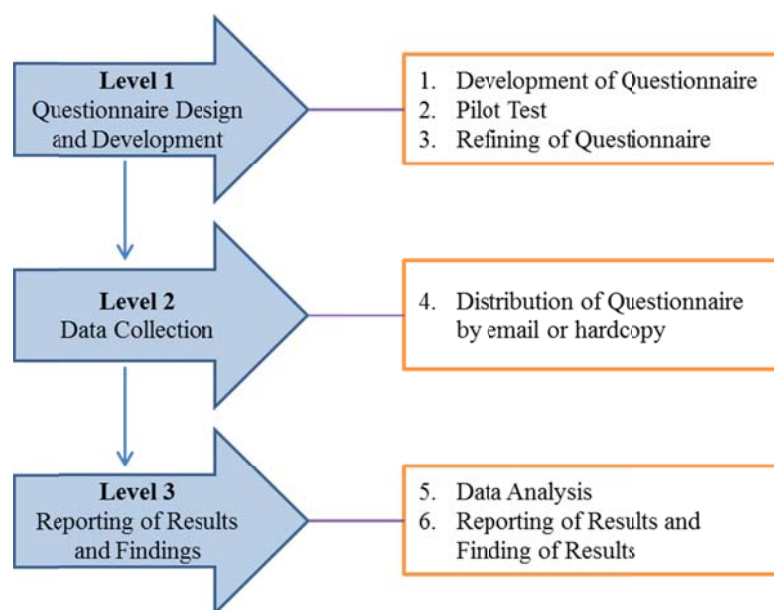
To ensure the reliability of the interview questions, face validity will be conducted. In short, the study design (quantitative, qualitative, and mixed methods) is aligned with the problem statement and research objectives.

3.3.1 Data Collection

This research incorporated parts of the data collection through the quantitative research with administrated structured questionnaire. The data are used to test the hypotheses developed for this research. The said data are collected

primarily from the self-administered questionnaire survey. English language is used to develop the questionnaire since it is the main communication medium in the oil and gas industry in Malaysia. Figure 3.1 shows the framework used to develop the survey process for this study. Level One is to develop a set of survey questionnaires and pre-test the survey questionnaire (pilot test). The questionnaire is modified based on the feedbacks from respondents and interviewees containing suggestions and opinions.

Figure 3.1: Framework for Developing Primary Data Collection



The first section of the questionnaire was to collect the details of the respondent's demographic details. The second section measured their general understanding and perception on visual management and the combination of performance and visual management. Respondents will rate their agreement with the statements based on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). The third section measured the impact and the effectiveness of visual management and the combination of performance and visual management in improving their operational and production performance. Similar to measurements of the second section, respondents will rate their agreement with the statements based on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

Level Two is related to the data collection process. The primary survey was conducted from 14th July until 25th Aug. Table 3.2 shows the target survey to be collected per week from 14th July until 25th July to achieve a target total minimum of 65 samples. The minimum target surveys comprises of the four department respondents – Operations, Subsurface, Drilling and Wells listed on Table 3.3

Table 3.2: Plan Survey to be Conducted Every Week

		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Plan Survey Per Week	0	5	13	18	13	10	6
Cum. Plan Survey	0	5	18	36	49	59	65

Table 3.3: Plan Survey Require by Department

Department	Target Survey
Operations	30
Offshore	15
Subsurface	10
Drilling	5
Wells	5

Level Three is the findings of the research. After the coding process and data entry, the consistency analysis was applied such as the frequency and cross-tabulation to make sure the data were free from any error. Statistical Package for the Social Sciences (SPSS) statistical package was used to prepare the findings of this research.

3.3.2 Sample

Purposive sampling, a non-probability sampling technique was used where subjects were selected based on their involvement on the implementation and experience they had prior and after the implementation of the tools. This is to

ensure the responses are reliably reflective of their experiences throughout the implementation process and to fulfil all the sample criteria. Saunders et al. (2016) and Greener (2008) prescribed the minimum sample size to be approximately 30. Furthermore, to further strengthen analysis results, Moore and McCabe (2012) said that, the rule of thumb to perform One-Sample t Test is to have a sample size of at least 40 or above, then the test can be safely used without regards to skewness or outliers. Hence, the minimum target sample required for this research is 65 sampling units, consisting 30 sampling units from Operations, 15 sampling units from Offshore, 10 sampling units from Subsurface, 5 from Wells and 5 from Drilling department as mentioned in Section 3.3.1.

Ultimately, choosing an appropriate sample size is very important as it can firstly avoid biases and more importantly, it is more representative of the target population.

3.3.3 Interview Instrument

The main reason of a qualitative research interview is to understand and gather feedback from the interviewees. An interview searches and addresses both factual and a meaning level to understand the feedback from the interviewees (Kvale, 1996). Usually, open-ended questions will be asked during the interview to seek and understand particularly the story behind the participant's experiences (McNamara, 1999). This will allow the interviewer to have an underlying in-depth information the impact of the implementation. Moreover, an interview is beneficial to follow up certain respondent's questionnaires or to further question the respondent's responses.

Interviews are usually easier for respondents, especially if a study searches for the views of a person or to understand how a person feels. In this research, the researcher will also conduct an interview with the employees who are the users of visual management. As a result of the interview, it will provide the researcher with a better understanding on both stand points, applying the visual management on their day-to-day work, why the tools can help them and how the tools can assist

them in their work. Furthermore, by conducting an interview, the researcher will be able to understand the perspectives or behaviours of the people impacted by the implementation of visual management. The approach taken for this research will be a standardized and open-ended interview. Such questions usually begin with a how, what, when, where, and why (such as "How visual management will help improve your work performance?"). The design of the questions should be easy to understand, straightforward and can be spoken distinctly and comprehensively. With this, it will help to aid a faster and more structured interview that in return will be easy to analyse and compared.

3.3.4 Interview Procedure

In preparation for the interview, the researcher will outline the procedure of the interview. Short and simple questions will be developed. A venue or setting with the least distraction will be chosen for the interview. Subsequently, a meeting will be booked based on the interviewee's schedule to ensure a specific timing is allocated for this interview. The researcher will be sensitive and patient towards any provocative and unconventional opinions. The researcher will also speak clearly and comprehensively and during the interview, the purpose will be explained and confidentiality will be addressed as well. To ensure a smooth interview takes place, the interviewee's expectation will be properly managed by indicating approximately how long the interview will take. The format of the interview will be explained to interviewee. Interviewee is allowed to clarify any doubts about the interview. Before the researcher starts asking about anything controversial, the interview will start off with factual questions. Fact-based questions will then be spread all the way through the interview. Questions about the future will be asked after the present question. Last but not least, the last question will allow respondents to add any other information on top of the interview question. Notes will be taken as a recording method for this interview.

There are three sections in questionnaire and first section question focus mainly to understand interviewee definition on performance. Second section of the questionnaire is to understand do scorecard helps to improve performance. If

the answer from respondents is yes, more details questions will be ask to understand why and how scorecard helps to improve performance. The last section is to find out is there any other performance issues in the organisation that the researcher can provide recommendation to improve the situation

3.4 Data Analysis

In this research, the researcher will use Statistical Package for the Social Science (SPSS) software version 20.0 as the analysis tool. The descriptive statistics (i.e. frequencies and means) are computed to measure the central tendencies and degree of variation in findings. Table 3.4 below shows the scores coded into SPSS for each point in the Likert scale. If the Mean value is below 3.0, it shows a negative tendency towards the particular item while Mean values above 3.0 indicate a positive tendency towards the item. Responses are also analysed by using the respondent management hierarchical levels for potential trends which are specific or unique to a certain management level.

Table 3.4: Likert scale score coding in SPSS

Score	Indicator
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strong Agree

3.4.1 Descriptive Statistics

In descriptive statistics, researchers attempt to describe the numerical characteristics of the data. To outline the sample including the standard deviation,

mean, or boxplot and explaining the characteristics of the probability distribution, the researcher will use descriptive statistics.

3.4.2 One-Sample t Test

In inferential statistics, the researchers will attempt to go beyond the data to generalise or predict the findings to the larger population (Johnson & Christensen, (2017) based on the characteristics of population. The data will be use to describe the population parameter (i.e. the population mean). The second area is to conduct a hypothesis testing. This is to test specific hypotheses about populations based on their sample data and to answer the research questions.

In this research, the researcher will use One-Sample t Test since it is suitable for a small sample size. The t -distribution is used since the sample size available for this research is moderate. For the use of a t test to be valid, the data should be normally distributed. Hence in this research, t test will be used to evaluate the lower and upper limit of the hypothesis base on the “Mean” minus the standard error mean with .05 or less error. If the probability is .05 or less, that means 5% or less of the sample will have error or mistake in claiming there is a variance between the pre and post-test scores in the implementation (Albrecht, n.d.). This test will also be used to determine the indication from the respondents whether they less agree, agree or strongly agree that the combination of performance and visual management will assist them to improve their operational performance.

Finally, this method is used to fulfil the requirement of the mixed model in which a research needs to be conducted prior and post the implementation to determine whether the combination of performance and visual management helps to further improve performance.

CHAPTER FOUR: RESEARCH RESULTS

4.1 Introduction

This chapter will include the discussions and present the findings from the respondents towards the perception of the combination of performance and visual management that will drive performance improvement. The main objective of this chapter is to analyse and present the findings from the collected data coming from the survey that was distributed to 95 employees and interviews with chosen employees from the selected multinational integrated Oil and Gas Company that is operating in Malaysia. In this chapter also, the researcher will discuss the findings from data analysis produced by SPSS that will be further explained with the presence of frequencies, cross tabulation and one-sample *t* test to meet the research objectives and hypotheses highlighted in Chapter 1. The qualitative research or the interview outcome will also be presented in this chapter.

4.2 Demographic Analysis

In this study, three demographic questions were included in Section A of the questionnaire to understand the respondents' profile. The questions covered respondent gender, department and position in the organisation. The data collected is subsequently analysed for descriptive statistics and frequencies.

To illustrate and describe the profile of respondents, descriptive analysis was used to differentiate their department, gender and position. A questionnaire was distributed to 95 selected integrated respondents from the Oil and Gas Company for this research. 91 questionnaires were returned and this translated to 95.8% response rate achieved. The criteria required to fill up the survey form was, the respondent's employment starts from 2008 or before and had experienced the entire process, prior and post implementation of visual management and scorecard – as explained in figure 3.1 and the user of scorecard. Hence, only selected

respondents from the 4 target departments were chosen to participate in the survey for this study.

The sample revealed that 75.8% of the respondents are male whereas the percentages of women are 24.2%. In the context of department, there are about 65.9% of respondents from Operations identified compared to 18.7%t from Subsurface, 6.6% from Drilling department and 8.8% from Wells department. As employee hierarchical levels might also influence their ratings of the study's variables, thus respondents were asked to indicate their position in the organisation based on three categories, namely Management (managerial or supervisory role), Engineer and Offshore Personnel. With respect to their position, the sample indicated that the majority of the respondents are as follow; 48.4% are engineer, 31.9% of the sample is management level, whereas only 19.8% of the sample is offshore personnel and this makes sense since the oil and gas industry is also predominantly a technically-oriented business. Last but not least, respondents' gender were also included in the demographic analysis since Male and Female see things differently and as result, they might have different perception or rating towards the study variables.

A summary of descriptive statistics of the respondents is shown in Table 4.1.

Table 4.1: Descriptive Statistics

	Count	Percent
Gender		
Male	69	75.8%
Female	22	24.2%
Department		
Operations	60	65.9%
Subsurface	17	18.7%
Drilling	6	6.6%
Wells	8	8.8%

Position		
Management	29	31.9%
Engineer	44	48.4%
Offshore Personnel	18	19.8%

4.3 Cross Tabulations Analysis

In this research, cross tabulations were used to jointly and collectively present the distribution of two or more variables. This will be done by tabulating the outcome one against the other in a two-dimensional grid. It uses a process from the multivariate frequency distribution of variables to create contingency tables to be presented in a matrix format. This will reveal the relationship between the two or more variables and as a result, the variables used to generate the crosstabs are correlated. It is also widely used to present the survey results to prove the interrelationship and interactions between variables.

4.3.1 Cross Tabulation Analysis for Hypothesis 1 against Department

In this research crosstabs were used to analyse the distribution of the respondents' demographic variable (i.e. department, gender and position) against the research hypothesis. The first cross tabulation table (Table 4.2) below which consist of an overall summary of the four departments differentiated by Operations, Subsurface, Drilling and Wells show the respondents' perception towards the first hypothesis - Visual Management will improve the operational performance. However, chi-square test is not performed due to the fact that there is at least 1 cell of the cross tabulation that has less than 5 counts. This will lead to invalid test statistics.

For the first statement, looking at the results of the analysis, the weightage is more to agree and above, and 58.2% of the respondent agreed that *Having a scorecard (visual management) helps to manage performance in general*. Moreover, 34.1% respondents strongly agreed that the use of scorecard (visual

management) helped them significantly in managing their operational performance. This is mainly due to the benefits of visual management which helps the employee to see a bigger picture, allowing them to structure their activities and providing them the ability to monitor and improve their performance. There is only 7.7% out of 100.0% of the respondents' who opted neutral, and neutral does not mean these 7 respondents, 4 from Operations, 2 from Subsurface and 1 from Wells department completely reject that visual management can help them in managing performance. All in all, the majority of the respondents from the various departments agreed with the statement.

For statement two, majority of the respondents regardless of departments strongly agreed (44.0% out of 100.0%) that *Having a scorecard (visual management) helps to provide better visibility that is required in managing performance* and only 4.4% or 4 respondents, 3 from Operations and 1 from Wells took a neutral stand. The modal frequencies (the highest percentages in each row) agreed with the statement. Looking at the total respondents, Operations department had the highest numbers respondents who agreed they experience the comprehensible and visibility through measurements and public display of information of the main production process from start to finish.

For statement three, we can see that 1 respondent from Subsurface that the individual disagreed by *Having a scorecard (visual management) helps to simplify performance management processes* which can be considered insignificant to totally object the ability of visual management to simplify performance management since it was only 1.1% out of 100%. Even if we combine the respondents who disagreed and were neutral 31.9%, it only sums up to 33.0% which is still considered less than 50% who disagreed that the ability of visual management to simplify the performance management process. Whereas, the combination of those who agreed and strongly agreed (40.7%+26.4%=67.1%) easily represents two-third of the majority who agreed that visual management can simplify the performance management process. For visual management to yield its full potential in improving performance, it must be linked to performance management which furnishes input into visual management.

For statement four, more than half of the respondents at 54.9% and almost all departments recorded a percentage of 50% and above agreeing ((Operations (51.7%), Drilling (47.1%) and Wells (87.5%)) that *Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement*. By that, it proves what was mentioned by Suzuki 1993, Visual Management will be the platform for continuous improvement. 16.5% of the respondents strongly agreed that scorecard (visual management) helps them to identify improvement opportunity, react to safety, quality, and efficiency problems and whereas, 26.4% of the respondents responded neutrally. Only 2.2% of the respondents, which translate to 1 from Operations and 1 from Subsurface disagreed that having a scorecard (visual management) DOES helps to identify ways to improve.

For statement five, only 1 respondent from Subsurface strongly disagreed and 2 respondents from Operations disagreed that *Having a daily scorecard for platform or FSO helps to improve the plant uptime*. 32 of the respondents, 19 from Operations, 9 from Subsurface, 1 from Drilling and 3 from Wells department, equivalent to 35.2%, had a neutral stand on this statement. But, the pie gets bigger where 45.1% of the respondents which translate to almost half of them with significant response mostly received from Drilling (83.3%), agreed with the statement. More importantly, 16.5% of the respondents strongly agreed that having a daily scorecard DOES help to improve the platform or FSO uptime.

For the last statement, *Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews*, no respondents strongly disagreed or disagreed. Only 26 respondents out of 91 respondents where 15 from Operations, 6 Subsurface, 2 Drilling and 3 Wells, responded neutrally. Whereas the majority agreed and strongly agreed with the ratio of 29:71 supporting that the daily scorecard does help the onshore support team to provide the support required by offshore. Referring back to literature review, it was mentioned that a scorecard (visual management) helps in providing job facilitation by offering a job overview, accurate and holistic understanding of their job requirement. In this case, daily scorecard provides a precise information

and understanding for onshore support team in providing their assistance to offshore team.

Table 4.1: Cross Tabulation Analysis for Hypothesis 1 against Department

	Hypothesis 1	Department	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to manage performance in general (<i>to test the 'does'</i>)	Operations	0	0	4 (6.7%)	35 (58.3%)	21(35.0%)	60 (100.0%)
		Subsurface	0	0	2 (11.8%)	6 (35.3%)	9 (52.9%)	17(100.0%)
		Drilling	0	0	0	5 (83.3%)	1 (16.7%)	6 (100.0%)
		Wells	0	0	1 (12.5%)	7 (87.5%)	0	8 (100.0%)
		Total	0	0	7 (7.7%)	53 (58.2%)	31 (34.1%)	91 (100.0%)
2	Having a scorecard (visual management) helps to provide better visibility that is required in managing performance (<i>to test the 'how'</i>)	Operations	0	0	3 (5.0%)	30 (50.0%)	27 (45.0%)	60 (100.0%)
		Subsurface	0	0	0	9 (52.9%)	8 (47.1%)	17 (100.0%)
		Drilling	0	0	0	2 (33.3%)	4 (66.7%)	6 (100.0%)
		Wells	0	0	1 (12.5%)	6 (75.0%)	1 (12.5%)	8 (100.0%)
		Total	0	0	4 (4.4%)	47 (51.6%)	40 (44.0%)	91 (100.0%)
3	Having a scorecard (visual management) helps to simplify performance management processes	Operations	0	0	19 (31.7%)	23 (38.3%)	18 (30.0%)	60 (100.0%)
		Subsurface	0	1 (5.9%)	6 (35.3%)	5 (29.4%)	5 (29.4%)	17 (100.0%)
		Drilling	0	0	2 (33.3%)	4 (66.7%)	0	6 (100.0%)
		Wells	0	0	2 (25.0%)	5 (62.5%)	1 (12.5%)	8 (100.0%)

		Total	0	1 (1.1%)	29 (31.9%)	37 (40.7%)	24 (26.4%)	91 (100.0%)
4	Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement (<i>to test the 'how'</i>)	Operations	0	1 (1.7%)	17 (28.3%)	31 (51.7%)	11 (18.3%)	60 (100.0%)
		Subsurface	0	1 (5.9%)	5 (29.4%)	8 (47.1%)	3 (17.6%)	17 (100.0%)
		Drilling	0	0	2 (33.3%)	4 (66.7%)	0	6 (100.0%)
		Wells	0	0	0	7 (87.5%)	1 (12.5%)	8 (100.0%)
		Total	0	2 (2.2%)	24 (26.4%)	50 (54.9%)	15 (16.5%)	91 (100.0%)
5	Having a daily scorecard for platform or FSO helps to improve the plant uptime	Operations	0	2 (3.3%)	19 (31.7%)	28 (46.7%)	11 (18.3%)	60 (100.0%)
		Subsurface	1 (5.9%)	0	9 (52.9%)	4 (23.5%)	3 (17.6%)	17 (100.0%)
		Drilling	0	0	1 (16.7%)	5 (83.3%)	0	6 (100.0%)
		Wells	0	0	3 (37.5%)	4 (50.0%)	1 (12.5%)	8 (100.0%)
		Total	1 (1.1%)	2 (2.2%)	32 (35.2%)	41 (45.1%)	15 (16.5%)	91 (100.0%)
6	Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews (<i>to test the 'how'</i>)	Operations	0	0	15 (25.0%)	30 (50.0%)	15 (25.0%)	60 (100.0%)
		Subsurface	0	0	6 (35.3%)	8 (47.1%)	3 (17.6%)	17 (100.0%)
		Drilling	0	0	2 (33.3%)	3 (50.0%)	1 (16.7%)	6 (100.0%)
		Wells	0	0	3 (37.5%)	4 (50.0%)	1 (12.5%)	8 (100.0%)
		Total	0	0	26 (28.6%)	45 (49.5%)	20 (22.0%)	91 (100.0%)

4.3.2 Cross Tabulation Analysis for Hypothesis 1 against Gender

Gender diversity can cause a difference in the result of the research outcome and this is mainly due to divergence of views between men and women from the management to engineer level. Obviously, males and females see things differently. Hence, to ensure both genders equally agreed with the hypothesis, a research was conducted to analyse the respondents' perception by gender. The cross tabulation table (Table 4.3) will show the responses for six statements according to male and female perception towards the first hypothesis - Visual Management will improve the operational performance.

For the first statement, looking at the results of the analysis, the responses were distributed almost equally between males and females. The minor variance between male and female respondents is due to the difference in total male and female respondents (69 male versus 22 female respondents). Still, despite the difference in the overall numbers of male and female respondents, the results still show that the majority of males (62.3%) and females (45.5%) agreed that *Having a scorecard (visual management) helps to manage performance in general*. The remaining 30.4% of male respondents and 45.5% of female respondents strongly agreed with the statement and only 7 respondents consisting of 5 males and 2 females took a neutral stand.

For the second statement, looking at the total point of view, we can see half of the population agreed that *Having a scorecard (visual management) helps to provide better visibility that is required in managing performance*. If we drill down to the details, we can see a higher percentage of females strongly agreeing with the statement as compared to males (54.5% female versus 40.6% male). Less than one quarter of the population had a neutral stand with the statement and more importantly only 1 female compared to 3 males had a neutral stand on this.

For the third statement, we can see only 1 male respondent disagreeing that *Having a scorecard (visual management) helps to simplify performance management processes*. Obviously, this is insignificant to conclude that having a scorecard DOES NOT help to simplify performance management process and the

researcher believe this is due to the respondent's job nature which hinders the individual from seeing the bigger picture. On the bright side, none of the female respondents disagreed with the statement. Instead, the majority (43.5%) of male respondents agreed with the statement and more than half of the female respondents responded higher than neutral when the total of those who agreed or strongly agreed were combined (31.8% agree and 31.8% strongly agree). Only the remaining 29 respondents consisting 21 males and 8 females had a neutral stand.

For the fourth statement, half of the population (56.5% males and 50% females) agreed that *Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement*. Furthermore, 16% of population, consisting of 17.4% males and 13.6% females strongly agreed with the statement. Only 24 respondents consisting 17 males and 7 females responded neutrally on the statement and only 1 male and 1 female disagreed that *Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement*.

For the fifth statement, 1 respondent strongly disagreed and 2 respondents and 3 of them were male respondents who disagreed that *Having a daily scorecard for platform or FSO helps to improve the plant uptime*. No female respondents strongly disagreed or disagreed which indirectly offset the 3 male respondents on the same scale. Moreover, almost half of the male population (49.3%) agreed with the statement. But, on the opposite side, majority of females (50.0%) had a neutral perception that *Having a daily scorecard for platform or FSO helps to improve the plant uptime*. As a researcher, I believe this due to the fact that all of the offshore personnel are male and hence practically the male respondent saw the benefits of the scorecard when they manage the plant and help reduce the downtime of the plant having the ability to see and identify the problem from the trending visual or indicators from the scorecard. More importantly, there are additional 15 respondents who strongly agreed and out of the 15 respondents, there were 4 female respondents. This is suffice to justify there were no biased respondents based on their gender.

Last but not least for the sixth statement, both genders almost equally opted to agree (49.3% male 50.0% female) that *Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews* despite quite a big number of males responded neutrally - 30.4%, whereas females who had a neutral stand were only 22.7%. Overall, we can conclude that having a daily scorecard DOES help onshore support teams to provide the necessary support required by offshore crews since two-third of the respondents agreed with the additional percentage of 20.3% males and 27.3% females strongly agreed with the statement.

Table 4.3: Cross Tabulation Analysis for Hypothesis 1 against Gender

	Hypothesis 1	Gender	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to manage performance in general (<i>to test the 'does'</i>)	Male	0	0	5 (7.2%)	43 (62.3%)	21 (30.4%)	69 (100.0%)
		Female	0	0	2 (9.1%)	10 (45.5%)	10 (45.5%)	22 (100.0%)
		Total	0	0	7 (7.7%)	53 (58.2%)	31 (34.1%)	91 (100.0%)
2	Having a scorecard (visual management) helps to provide better visibility that is required in managing performance (<i>to test the 'how'</i>)	Male	0	0	3 (4.3%)	38 (55.1%)	28 (40.6%)	69 (100.0%)
		Female	0	0	1 (4.5%)	9 (40.9%)	12 (54.5%)	22 (100.0%)
		Total	0	0	4 (4.4%)	47 (51.6%)	40 (44.0%)	91 (100.0%)
3	Having a scorecard (visual management) helps to simplify performance management processes	Male	0	1 (1.4%)	21 (30.4%)	30 (43.5%)	17 (24.6%)	69 (100.0%)
		Female	0	0	8 (36.4%)	7 (31.8%)	7 (31.8%)	22 (100.0%)
		Total	0	1	29 (31.9%)	37 (40.7%)	24 (26.4%)	91 (100.0%)
4	Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement (<i>to test the 'how'</i>)	Male	0	1 (1.4%)	17 (24.6%)	39 (56.5%)	12 (17.4%)	69 (100.0%)
		Female	0	1 (4.5%)	7 (31.8%)	11 (50.0%)	3 (13.6%)	22 (100.0%)
		Total	0	2 (2.2%)	24 (26.4%)	50 (54.9%)	15 (16.5%)	91 (100.0%)

5	Having a daily scorecard for platform or FSO helps to improve the plant uptime	Male	1 (1.4%)	2 (2.9%)	21 (30.4%)	34 (49.3%)	11 (15.9%)	69 (100.0%)
		Female	0	0	11 (50.0%)	7 (31.8%)	4 (18.2%)	22 (100.0%)
		Total	1 (1.1%)	2 (2.2%)	32 (35.2%)	41 (45.1%)	15 (16.5%)	91 (100.0%)
6	Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews (<i>to test the 'how'</i>)	Male	0	0	21 (30.4%)	34 (49.3%)	14 (20.3%)	69 (100.0%)
		Female	0	0	5 (22.7%)	11 (50.0%)	6 (27.3%)	22 (100.0%)
		Total	0	0	26 (28.6%)	45 (49.5%)	20 (22.0%)	91 (100.0%)

4.3.3 Cross Tabulation Analysis for Hypothesis 1 against Position

Position can also be one of the factors impacting the research outcome. This is mainly due to the level of awareness on performance management, visual management and scorecard from the selected sample. Most of the time, management level employees are well exposed to the performance and visual management tools and processes whereas, engineers and offshore personnel job scopes tend to be more focused and exposed more on technical work. As a result, engineers and offshore personnel might feel sceptical on the ability of these processes and tools to help improve performance. Hence, the cross-tabulation table (Table 4.4) will show the responses for six statements based on positions, towards the first hypothesis - Visual Management will improve the operational performance.

For the first statement, looking at the results of the analysis, there is no neutral response from managerial-level employees. Only 7 respondents from a total of 91 respondents took a neutral stand – 5 respondents were engineers and 2 offshore personnel which is pretty consistent with the assumption the researcher mentioned just now. Whereas on the flip side, majority of the respondents - 53% regardless of position agreed that *Having a scorecard (visual management) helps to manage performance in general*. Furthermore, it was further strengthened by an additional 34.1%; 9 respondents from Management, 16 respondents from Engineer and 6 respondents from Offshore Personnel who strongly agreed with the statement.

For the second statement, no management-level respondents responded neutrally or disagreed or strongly disagreed. Only 4 respondents, 3 engineers and 1 offshore personnel felt neutral that *Having a scorecard (visual management) helps to provide better visibility that is required in managing performance*. The rest of the respondents with the total of 87 respondents, consisting of 47 respondents agreed and 40 respondents strongly agreed. The result also shows that management-level completely support the statement since none of them answered neutral or below. The distribution between those who agreed and strongly agreed are almost 50:50 except for offshore personnel as the gap is slightly bigger and

this is mainly due to the total numbers of offshore personnel which was smaller compared to engineers and employees from the management level.

For the third statement, again, only 1 engineer responded disagree that *Having a scorecard (visual management) helps to simplify performance management processes*. As a researcher, again, I strongly believe that this is mainly due to the respondent's job description – the respondent is an engineer hence, there is a very high chance that the respondent's portfolio does not involve performance management process. Hence, that hinders the respondent from responding accurately. Almost one third of the respondents reacted to a neutral stand on the statement. On the hindsight, almost two third of the respondents replied agreed and strongly agreed. Nevertheless, the overall results still showed a heavier weightage of respondents who agreed with the statement compared to others.

For the fourth statement, we have 2 engineers who disagreed that *Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement*. This is offset by 52.3% where majority of the engineers agreed that having a scorecard (visual management) DOES helps to identify ways to improve by providing the information that is required for improvement. We can also conclude that majority of the management and offshore personnel employee also agreed with the statement since the rate of respondents who agreed amounted to 58.6% from management and 55.6% from offshore personnel. This was also further strengthened with an additional 16.5% who strongly agreed. Despite the percentage of those who strongly agreed is not as big compared to those who agreed and had a neutral stand, but if we combine those agreed and strongly agreed, it will derive to a total of 71.4% and that is sufficient to represent the majority of the respondents.

For statement five, 1 engineer strongly disagreed whereas 1 engineer and 1 offshore personnel disagreed that *Having a daily scorecard for platform or FSO helps to improve the plant uptime*. The 3 respondents were not even close to one-third of the total respondents. Hence, 3 respondents could not even represent the minority respondents who disagreed with the statement. 35.2% of the respondents

consisting of 9 respondents from the management, 18 engineers and 5 offshore personnel opted to be neutral. Almost two-third of the remaining respondents agreed and strongly agreed that having a daily scorecard for platform or FSO DOES helps to improve the plant uptime. The two third of the respondents consists 45.1% that answered agreed and 16.5% picked strongly agreed with the statement.

Last but not least for the sixth statement, almost half of the total respondents (45.1%) agreed that *Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews*. Moreover, 54.5% of the total engineer respondents agreed with the statement. This is very important since one of the objectives of having a daily scorecard is to help the onshore engineers to provide the necessary support required by the offshore team. This is also supported by 38.9% who agreed and 38.9% who strongly agreed which were responded by offshore personnel. This is good enough to provide the evidence that the offshore team received suitable support from the onshore team after daily scorecard implementation. A positive feedback was also received from the management where 48.3% of them agreed and 20.7% strongly agreed that the daily scorecard does help the onshore and offshore team work better as a team.

Table 4.4: Cross Tabulation Analysis for Hypothesis 1 against Position

	Hypothesis 1	Position	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to manage performance in general (<i>to test the 'does'</i>)	Management	0	0	0	20 (69.0%)	9 (31.0%)	29 (100.0%)
		Engineer	0	0	5 (11.4%)	23 (52.3%)	16 (36.4%)	44 (100.0%)
		Offshore Personnel	0	0	2 (11.1%)	10 (55.6%)	6 (33.3%)	18 (100.0%)
		Total	0	0	7 (7.7%)	53 (58.2%)	31 (34.1%)	91 (100.0%)
2	Having a scorecard (visual management) helps to provide better visibility that is required in managing performance (<i>to test the 'how'</i>)	Management	0	0	0	15 (51.7%)	14 (48.3%)	29 (100.0%)
		Engineer	0	0	3 (6.8%)	22 (50.0%)	19 (43.2%)	44 (100.0%)
		Offshore Personnel	0	0	1 (5.6%)	10 (55.6%)	7 (38.9%)	18 (100.0%)
		Total	0	0	4 (4.4%)	47 (51.6%)	40 (44.0%)	91 (100.0%)
3	Having a scorecard (visual management) helps to simplify performance management processes	Management	0	0	10 (34.5%)	14 (48.3%)	5 (17.2%)	29 (100.0%)
		Engineer	0	1 (2.3%)	16 (36.4%)	16 (36.4%)	11 (25.0%)	44 (100.0%)
		Offshore Personnel	0	0	3 (16.7%)	7 (38.9%)	8 (44.4%)	18 (100.0%)
		Total	0	1 (1.1%)	29 (31.9%)	37 (40.7%)	24 (26.4%)	91 (100.0%)

4	Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement (<i>to test the 'how'</i>)	Management	0	0	9 (31.0%)	17 (58.6%)	3 (10.3%)	29 (100.0%)
		Engineer	0	2 (4.5%)	12 (27.3%)	23 (52.3%)	7 (15.9%)	44 (100.0%)
		Offshore Personnel	0	0	3 (16.7%)	10 (55.6%)	5 (27.8%)	18 (100.0%)
		Total	0	2 (2.2%)	24 (26.4%)	50 (54.9%)	15 (16.5%)	91 (100.0%)
5	Having a daily scorecard for platform or FSO helps to improve the plant uptime	Management	0	0	9 (31.0%)	15 (51.7%)	5 (17.2%)	29 (100.0%)
		Engineer	1 (2.3%)	1 (2.3%)	18 (40.9%)	16 (36.4%)	8 (18.2%)	44 (100.0%)
		Offshore Personnel	0	1 (5.6%)	5 (27.8%)	10 (55.6%)	2 (11.1%)	18 (100.0%)
		Total	1 (1.1%)	2 (2.2%)	32 (35.2%)	41 (45.1%)	15 (16.5%)	91 (100.0%)
6	Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews (<i>to test the 'how'</i>)	Management	0	0	9 (31.0%)	14 (48.3%)	6 (20.7%)	29 (100.0%)
		Engineer	0	0	13 (29.5%)	24 (54.5%)	7 (15.9%)	44 (100.0%)
		Offshore Personnel	0	0	4 (22.2%)	7 (38.9%)	7 (38.9%)	18 (100.0%)
		Total	0	0	26 (28.6%)	45 (49.5%)	20 (22.0%)	91 (100.0%)

4.3.4 Cross Tabulation Analysis for Hypothesis 2 against Department

In this section, the cross tabulation table (Table 4.5) below will show the analysis of cross tabulation for hypothesis 2 (Performance Management hybrid with Visual Management will further improve performance) against Department of the respondents.

For the first statement, looking at the results of the analysis, there is a total 5 respondents; 1 disagreed and 4 were neutral that *Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals*. Surprisingly, the 4 respondents were from Operations department. The researcher strongly believed again, this might be due to the nature of respondents' work that might hinder them from being able to see a greater picture especially if the respondents were too focused on the technical side. The majority (50.5%) of the respondents agreed and this consists of 29 respondents from Operations department, 6 respondents from Subsurface, 5 from Drilling and 6 from Wells. This is further reinforced by 44.0% of the respondents who strongly agreed in which 26 respondents were from Operations department, 11 from Subsurface, 1 from Drilling and 2 from Wells. The combination of respondents from Operations who agreed and strongly agreed (29 respondents "Agree" + 26 respondents "Strongly Agree") can easily offset the 5 respondents who disagreed and chose to be neutral.

For the second statement, only 5 out of 91 respondents disagreed that *Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance*. These 5 respondents, 4 of them are from Operations department and 1 of them is from Wells. Despite 26.4% of them chose to be neutral with the statement, 46.2% agreed and the remaining 22.0% strongly agreed. Even though zero respondents from Wells and Drilling strongly agreed but the result is still good enough to show that the majority of respondents from these two departments agreed with the statement.

For the third statement, again, only 1 respondent from Subsurface department strongly disagreed and 1 respondent from Operations disagreed that *Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)*. In contrast, 58.8% of the total respondents from Subsurface and 56.7% of the total respondents from Operations agreed with the statement. This is further solidified by an additional 12 respondents from Operations department and 2 from Subsurface who strongly agreed with the ability of the daily scorecard to help them meet the requirement or target of the Reservoir Management Plan (RMP). For the Drilling department, majority of the them (83.3%) agreed with the statement whereas for Wells, 37.5% had neutral stand, 37.5% agreed and 25.0% strongly agreed with the statement.

For the fourth statement, only 1 Operations department respondent disagreed that *Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)* and only additional 5 respondents from Operations department had a neutral stand. Whereas the majority which is 66.7% of Operations department respondents agreed and 23.3% strongly agreed. With this, we can consider a vast majority 90% (66.7% agreed + 23.3% strongly agreed) from Operations department supported that having a daily scorecard DOES help them to manage their facility performance. For Subsurface, only 4 respondents and 3 respondents from Wells had a neutral stand - no Drillers responded neutrally. There are sufficient respondents from Subsurface, Drilling and Wells department who agreed and strongly agreed to offset the neutral respondents. The majority of Subsurface department respondents - 52.9% strongly agreed, 83.3% from Drilling and 37.5% from Wells. Despite Wells having only 37.5% respondents who agreed, they have an additional 25.0% who strongly agreed and this is enough to sum it up to $62.5\% = 37.5\% + 25.0\%$ of Wells respondents responded above neutral.

For statement five, none strongly disagreed and disagreed that *Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals*. Only a total of 12 respondents 7 from Operations department, 2 from Subsurface and 3 from Drilling responded

neutrally. On the contrary, two third of the respondents from Subsurface Department - 58.8% agreed with the statement whereas, 46.7% from Operations, 33.3% from Drilling and 50.0% from Wells agreed that having a daily scorecard for platform or FSO DOES help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals. To further support this assumption, an additional 35 out of 91 respondents consisting 41% from Operations, 29.4% from Subsurface, 66.7% from Drilling and 12.5% from Wells picked strongly agreed.

Table 4.5: Cross Tabulation Analysis for Hypothesis 2 against Department

	Hypothesis 2	Department	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals	Operations	0	1 (1.7%)	4 (6.7%)	29 (48.3%)	26 (43.3%)	60 (100.0%)
		Subsurface	0	0	0	6 (35.3%)	11 (64.7%)	17 (100.0%)
		Drilling	0	0	0	5 (83.3%)	1 (16.7%)	6 (100.0%)
		Wells	0	0	0	6 (75.0%)	2 (25.0%)	8 (100.0%)
		Total	0	1 (1.1%)	4 (4.4%)	46 (50.5%)	40 (44.0%)	91 (100.0%)
2	Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance (to test what is the impact having a tools without the combination of performance management is visual management or vice versa)	Operations	0	4 (6.7%)	13 (21.7%)	28 (46.7%)	15 (25.0%)	60 (100.0%)
		Subsurface	0	0	7 (41.2%)	5 (29.4%)	5 (29.4%)	17 (100.0%)
		Drilling	0	0	3 (50.0%)	3 (50.0%)	0	6 (100.0%)
		Wells	0	1 (12.5%)	1 (12.5%)	6 (75.0%)	0	8 (100.0%)
		Total	0	5 (5.5%)	24 (26.4%)	42 (46.2%)	20 (22.0%)	91 (100.0%)

3	Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)	Operations	0	1 (1.7%)	13 (21.7%)	34 (56.7%)	12 (20.0%)	60 (100.0%)
		Subsurface	1 (5.9%)	0	4 (23.5%)	10 (58.8%)	2 (11.8%)	17 (100.0%)
		Drilling	0	0	1 (16.7%)	5 (83.3%)	0	6 (100.0%)
		Wells	0	0	3 (37.5%)	3 (37.5%)	2 (25.0%)	8 (100.0%)
		Total	1 (1.1%)	1 (1.1%)	21 (23.1%)	52 (57.1%)	16 (17.6%)	91 (100.0%)
4	Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)	Operations	0	1 (1.7%)	5 (8.3%)	40 (66.7%)	14 (23.3%)	60 (100.0%)
		Subsurface	0	0	4 (23.5%)	9 (52.9%)	4 (23.5%)	17 (100.0%)
		Drilling	0	0	0	5 (83.3%)	1 (16.7%)	6 (100.0%)
		Wells	0	0	3 (37.5%)	3 (37.5%)	2 (25.0%)	8 (100.0%)
		Total	0	1 (1.1%)	12 (13.2%)	57 (62.6%)	21 (23.1%)	91 (100.0%)
5	Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals	Operations	0	0	7 (11.7%)	28 (46.7%)	25 (41.7%)	60 (100.0%)
		Subsurface	0	0	2 (11.8%)	10 (58.8%)	5 (29.4%)	17 (100.0%)
		Drilling	0	0	0	2 (33.3%)	4 (66.7%)	6 (100.0%)
		Wells	0	0	3 (37.5%)	4 (50.0%)	1 (12.5%)	8 (100.0%)
		Total	0	0	12 (13.2%)	44 (48.4%)	35 (38.5%)	91 (100.0%)

4.3.5 Cross Tabulation Analysis for Hypothesis 2 against Gender

In this section, the cross tabulation table (Table 4.6) below will show the analysis of cross tabulation for hypothesis 2 (Performance Management hybrid with Visual Management will improve performance) against Gender of the respondents.

For the first statement, looking at the results of the analysis, only 1 male respondent disagreed but no females disagreed that *Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals*. Moreover, only 4 respondents out of 91 respondents responded neutrally on the statement which consists of 3 males and 1 female respondent. These are easily offset by the 50.5% of the respondents, consisting of 53.9% male respondents and 40.9% female respondents who agreed with the statement. An additional 44.0% of the respondents consisting of 40.6% males and 54.5% females strongly agreed with the statement and this has further strengthened the evidence to support the statement of having a scorecard (visual management) DOES helps to provide organization the clarity on performance in comparison to business unit goals.

For the second statement, less than one-third of the majority respondents disagreed with the statement *Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance* and these 5 respondents were males. The pie gets bigger when it comes to those who were neutral where 20 respondents, 17 males and 7 females had a neutral stand on the statement. But eventually, almost two-third of the total respondents consisting of 50.7% males and 31.8% females agreed with the statement. The remaining 20 respondents, 12 males and 8 females strongly agreed with the statement and this is sufficient to conclude that visual management DOES help performance management to achieve its full potential if we integrate both tools and processes together.

For the third statement, again, 2 male respondents, 1 strongly disagreed and 1 disagreed that *Having a daily scorecard for platform helps to meet the*

requirement or target for Reservoir Management Plan (RMP). Despite 21 respondents – 16 males and 5 females responded neutral stands, this was offset by more than half of the total respondents (52 respondents) with majority males respondents - 56.5% opted agreed with the statement. Furthermore, 12 males and 4 females strongly agreed that having a daily scorecard for platform DOES helps to meet the requirement or target for Reservoir Management Plan (RMP).

For the fourth statement, only 1 male respondent disagreed with the additional 8 males and 4 females who were neutral on their perception of *Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)*. We can see a significant jump on the respondent's perception among those who agreed with the statement from 14.1% including disagree to 62.6% and the total for both gender to be above 50% who agreed with the statement. Not to mention, an additional 23.1% of the respondents involved were 15 males and 7 females who strongly agreed with the statement. All in all, two-third or majority of the respondent actually supported that having a daily scorecard for platform DOES help to manage Production, Gas Lift Injection, Water Injection performance.

For the fifth statement, none of the males and females strongly disagreed and disagreed that *Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals*. Less than one-third of the total respondents, 10 males and 2 females responded neutrally on the statement. On the positive side, we can see 47.8% males and 50% females agreed with the statement. This was further supported by an additional 37.7% male and 40.9% females who strongly agreed with the statement. As a result, we can also see the ratio from neutral to strongly agree category at almost 50:50 between male and female despite the total male respondents was 3 times higher than that of the females.

Table 4.6: Cross Tabulation Analysis for Hypothesis 2 against Gender

	Hypothesis 2	Gender	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals	Male	0	1 (1.4%)	3 (4.3%)	37 (53.9%)	28 (40.6%)	69 (100.0%)
		Female	0	0	1 (4.5%)	9 (40.9%)	12 (54.5%)	22 (100.0%)
		Total	0	1 (1.1%)	4 (4.4%)	46 (50.5%)	40 (44.0%)	91 (100.0%)
2	Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance (to test what is the impact having a tools without the combination of performance management is visual management or vice versa)	Male	0	5 (7.2%)	17 (24.6%)	35 (50.7%)	12 (17.4%)	69 (100.0%)
		Female	0	0	7 (31.8%)	7 (31.8%)	8 (36.4%)	22 (100.0%)
		Total	0	5 (5.5%)	24 (26.4%)	42 (46.2%)	20 (22.0%)	91 (100.0%)
3	Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)	Male	1 (1.4%)	1 (1.4%)	16 (23.2%)	39 (56.5%)	12 (17.4%)	69 (100.0%)
		Female	0	0	5 (22.7%)	13 (59.1%)	4 (18.2%)	22 (100.0%)
		Total	1 (1.1%)	1 (1.1%)	21 (23.1%)	52 (57.1%)	16 (17.6%)	91 (100.0%)
4	Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)	Male	0	1 (1.4%)	8 (11.6%)	45 (65.2%)	15 (21.7%)	69 (100.0%)
		Female	0	0	4 (18.2%)	12 (54.5%)	6 (27.3%)	22 (100.0%)
		Total	0	1 (1.1%)	12 (13.2%)	57 (62.6%)	21 (23.1%)	91 (100.0%)

5	Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals	Male	0	0	10 (14.5%)	33 (47.8%)	26 (37.7%)	69 (100.0%)
		Female	0	0	2 (9.1%)	11 (50.0%)	9 (40.9%)	22 (100.0%)
		Total	0	0	12 (13.2%)	44 (48.4%)	35 (38.5%)	91 (100.0%)

4.3.6 Cross Tabulation Analysis for Hypothesis 2 against Position

In this section, the cross tabulation table (Table 4.7) below will show the analysis of cross tabulation for hypothesis 2 (Performance Management hybrid with Visual Management will improve performance) against Position of the respondents.

For the first statement, the result shows only 1 management-level respondent disagreeing that *Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals*. Meanwhile, 4 respondents, 2 management and 2 engineers but 0 Offshore Personnel had a neutral stand on the statement. However, majority, be it Management or Engineer or Offshore Personnel, agreed with the statement and this is supported by the analysis results where 55.2% Management, 43.2% Engineer and 61.1% Offshore personnel selected the respective answer. Adding on, it was further strengthened by the additional 44.0% respondents with a majority of engineers (52.3%) who strongly agreed with the statement.

For the second statement, only 3 engineers and 2 offshore personnel disagreed that *Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance*. The good news is no management respondents disagreed with the statement although 7 management respondents were neutral which was not discouraging. Likewise, 12 respondents among the Engineers and Offshore Personnel were neutral but this was offset by almost half of the total number of engineer respondents - 40.9% and more than one-third of the offshore personnel - 33.3% agreed with the statement. Furthermore, 62.1% of management respondents agreed with the statement too. The remaining 20 respondents helped to strengthen the evidence to prove that *Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance* when they strongly agreed with the statement.

For the third statement, 1 Engineer strongly disagreed and 1 Offshore Personnel disagreed that *Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)*. We can also see that 7 respondents from Management, 11 from Engineers and 3 from Offshore Personnel were neutral on the statement. But on the flipside, we can see majority of them regardless of their position – in which 58.6% from Management, 59.1% from Engineers and 50.0% from Offshore Personnel agreed with the statement. With an additional 16 respondents that consist of 5 from Management, 6 Engineers and 5 Offshore Personnel responded strongly agreed with the statement. It is now proven that having a daily scorecard for platform DOES helps to meet the requirement or target for Reservoir Management Plan (RMP).

For the fourth statement, 57 respondents equal to 62.6% of the total respondents, agreed that *Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)*. Only 12 respondents responded neutrally and those were considered the minimum from management, engineer and offshore personnel. No respondent from Management and Engineer disagreed except 1 Offshore Personnel who disagreed and obviously this is not a significant percentage to represent the majority. Moreover, 21 respondents who strongly agreed that having a daily scorecard for platform DOES help to manage Production, Gas Lift Injection, Water Injection performance and this is adequate to conclude that majority of the respondents agreed with the statement.

For the fifth statement, no respondents strongly disagreed or disagreed and meanwhile only 12 respondents comprised of 1 Management, 9 Engineers and 2 Offshore Personnel were neutral on *Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals*. The majority of the respondents agreed and strongly agreed as we can see 48.4% of the respondents agreed and 38.5% strongly agreed. We can also see the responses by managements, engineers and offshore personnel to be almost equally distributed.

Table 4.7: Cross Tabulation Analysis for Hypothesis 2 against Position

	Hypothesis 2	Position	1=Strongly Disagree Count (%)	2=Disagree Count (%)	3=Neutral Count (%)	4=Agree Count (%)	5=Strongly Agree Count (%)	Total Count (%)
1	Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals	Management	0	1 (3.4%)	2 (6.9%)	16 (55.2%)	10 (34.5%)	29 (100.0%)
		Engineer	0	0	2 (4.5%)	19 (43.2%)	23 (52.3%)	44 (100.0%)
		Offshore Personnel	0	0	0	11 (61.1%)	7 (38.9%)	18 (100.0%)
		Total	0	1 (1.1%)	4 (4.4%)	46 (50.5%)	40 (44.0%)	91 (100.0%)
2	Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance (to test what is the impact having a tools without the combination of performance management is visual management or vice versa)	Management	0	0	7 (24.1%)	18 (62.1%)	4 (13.8%)	29 (100.0%)
		Engineer	0	3 (6.8%)	12 (27.3%)	18 (40.9%)	11 (25.0%)	44 (100.0%)
		Offshore Personnel	0	2 (11.1%)	5 (27.8%)	6 (33.3%)	5 (27.8%)	18 (100.0%)
		Total	0	5 (5.5%)	24 (24.6%)	42 (46.2%)	20 (22.0%)	91 (100.0%)
3	Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)	Management	0	0	7 (24.1%)	17 (58.6%)	5 (17.2%)	29 (100.0%)
		Engineer	1 (2.3%)	0	11 (25.0%)	26 (59.1%)	6 (13.6%)	44 (100.0%)
		Offshore Personnel	0	1 (5.6%)	3 (16.7%)	9 (50.0%)	5 (27.8%)	18 (100.0%)

		Total	1 (1.1%)	1 (1.1%)	21 (23.1%)	52 (57.1%)	16 (17.6%)	91 (100.0%)
4	Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)	Management	0	0	1 (3.4%)	21 (72.4%)	7 (24.1%)	29 (100.0%)
		Engineer	0	0	9 (20.5%)	26 (59.1%)	9 (20.5%)	44 (100.0%)
		Offshore Personnel	0	1 (5.6%)	2 (11.1%)	10 (55.6%)	5 (27.8%)	18 (100.0%)
		Total	0	1 (1.1%)	12 (13.2%)	57 (62.6%)	21 (23.1%)	91 (100.0%)
5	Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals	Management	0	0	2 (6.9%)	16 (55.2%)	11 (37.9%)	29 (100.0%)
		Engineer	0	0	9 (20.5%)	20 (45.5%)	15 (34.1%)	44 (100.0%)
		Offshore Personnel	0	0	1 (5.6%)	8 (44.4%)	9 (50.0%)	18 (100.0%)
		Total	0	0	12 (13.2%)	44 (48.4%)	35 (38.5%)	91 (100.0%)

4.4 One-Sample t Test Analysis

One-sample t test is designed to test whether the mean of a distribution differs significantly from some pre-set value and most of the time this test is used to compare the sample means. In this research, the researcher used One-sample t test to determine whether there is sufficient evidence to conclude the mean of the population from the sample drawn from a defined population.

4.4.1 One-Sample t Test Analysis for Hypothesis 1

In this section, (Table 4.8) below will show the analysis of One-sample t test table for hypothesis 1 (Visual Management will improve the operational performance). The One-sample t test analysis indicates the Mean for *Having a daily scorecard for platform or FSO helps to improve the plant uptime* is 3.74 which is the lowest among all other statements for Hypothesis 1. Not only the Mean is the lowest for this statement, the Standard Deviation .800, is also the highest among all. This indicates that the deviation of respondents' answers is high. It was strongly believed this is mainly due to the perception and interpretation that varied from each respondent about the statement. Some of the respondents might have perceived that only tangible item can improve the plant or facility uptime (i.e. a brand new pump/compressor or a more frequent preventive maintenance) instead of a piece of scorecard or report. But they fail to realise that the scorecard is the one that helps to facilitate the conversation to generate the next course of action to optimize production. The lower limit for this statement ($= \text{Mean} - (1.96 * \text{Std Error Mean})$) shows that 3.57 is the lowest overall average respondents whereas the upper limit ($= \text{Mean} + (1.96 * \text{Std Error Mean})$) show the highest is 3.90. The "1.96" is the critical value for 95 percent confidence interval. This concludes that the Mean is less than 4 hence it indicates that the respondents did not really agree with the statement.

For the second statement, the Mean is 3.86, for *Having a scorecard (visual management) helps to identify ways to improve by providing the information that required for improvement*, +0.12 higher than the first statement. The Std. Deviation is .708 which still considered high since there were 2 respondents who

answered disagreed. Moreover, we can see the lower limit shows 3.71 and this is mainly due to quite a number of respondents who responded to a neutral stand for this statement. As a researcher, I strongly believe this is mainly due to the nature of the respondents' job and exposure that restricted them from realising the benefit or value added for this tool. On the positive side, the Upper Limit nicely hit 4.00 which serves to prove and conclude that the Mean is 4 and that indicates that the respondents at least agreed with the statement.

For the third statement, the Mean is 3.92, for *Having a scorecard (visual management) helps to simplify performance management processes*, +0.06 higher than the second statement. Despite the fact that the Mean is getting higher, the Lower Limit is 3.76 which is still considered low. This can be contributed by the higher Std. Deviation .792. No doubt we see a high deviation of answer from the respondents but this can be addressed by the organisation if they provide more education to the employees to ensure they know how to use the tools and are able to see the benefits of the tools. Despite all that, the Upper Limit now is 4.09 which is above 4. With that, we can conclude the Mean is 4 and it shows that the respondents at least agreed with the statement.

For the fourth statement, the Mean is 3.93 for *Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews*, +0.01 higher than the third statement. We can also see improvement on the Std. Deviation from .792 to .712 for statement four. With that, it reduced the gap of the Lower Limit vs Upper Limit. The Lower Limit now is 3.79 and the Upper Limit is 4.08. Although the Upper Limit is 0.01 lower compared to the third statement, we are still able to conclude that the Mean is 4 and hence it shows that the respondents at least agreed with the statement.

For statement five, the Mean is 4.26 for *Having a scorecard (visual management) helps to manage performance in general*, a significant increase of +0.33 compared to the previous statement. This Mean result also translates to a lower Std. Deviation .593, -0.119 reduction compared to the previous statement's Std. Deviation. This shows that the variance or the gap of respondents' answer is growing smaller. For the Lower Limit, the result showed 4.14 which is above 4

and the Upper Limit showed 4.39. It is conclusive that the Mean is above 4, therefore respondents agreed with the statement. This result supports the theory of Liff and Posey (2004) that Visual Management leads to improving poor performance and sustaining desired behaviours.

For statement six, the Mean is 4.40 for *Having a scorecard (visual management) helps to provide better visibility that is required in managing performance* which is the highest among all the statements for hypothesis 1, +0.66 improvements compared to the lowest Mean. We can also see the lowest Std. Deviation, .575 among all the statements and it was -.225 reductions as compared to the highest Std. Deviation in hypothesis 1. As a result of the higher Mean and lower Std. Deviation, we can see both Lower Limit and Upper Limit results more than 4.20. It can be inferred that Mean is more than 4 and it indicates respondents absolutely agreed with the statement.

Table 4.8: One Sample T-Test Result for Hypothesis 1

	Null Hypothesis	N	Mean	Std. Deviation	Std. Error Mean	95% CI Lower Limit	95% CI Upper Limit	Conclusion mean is
H1	Having a daily scorecard for platform or FSO helps to improve the plant uptime	91	3.74	.800	.084	3.57	3.90	Mean is less than 4, indicating the respondent less agree
H1	Having a scorecard (visual management) helps to identify ways to improve by providing the information that required for improvement (to test the 'how')	91	3.86	.708	.074	3.71	4.00	Mean is 4, indicating the respondent at least agreed with the statement
H1	Having a scorecard (visual management) helps to simplify performance management processes	91	3.92	.792	.083	3.76	4.09	Mean is 4, indicating the respondent at least agreed with the statement
H1	Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews (to test the 'how')	91	3.93	.712	.075	3.79	4.08	Mean is 4, indicating the respondent at least agreed with the statement
H1	Having a scorecard (visual management) helps to manage performance in general (to test the 'does')	91	4.26	.593	.062	4.14	4.39	Mean is more than 4, indicating the respondent agreed with the statement
H1	Having a scorecard (visual management) helps to provide better visibility that is required in managing performance (to test the 'how')	91	4.40	.575	.060	4.28	4.51	Mean is more than 4, indicating the respondent agreed with the statement

4.4.2 One-Sample t Test Analysis for Hypothesis 2

In this section, (Table 4.9) below will show the analysis of One-sample t test table for hypothesis 2 (Performance Management hybrid with Visual Management will further improve performance). The One-sample t test analysis for Hypothesis 2 indicates the Mean for *Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance* is 3.85 which is the lowest among all other statements for Hypothesis 2. Furthermore, the Std. Deviation result is not encouraging too. The analysis result also showed the Std. Deviation is .829, the highest compared to others. All these were mainly contributed by the respondents' perception especially those who had been serving long enough in the industries. According to some of the respondents, they were curious as to how Performance Management cannot achieve their full potential since Performance Management framework had existed in the industries for quite some time and furthermore those top 5 Integrated Oil and Company are still practicing the same process from the same framework. For the Lower and Upper Limit, it can be seen there is gap between those numbers, partly contributed by a higher Std. Deviation. The Lower Limit result showed 3.68 and Upper Limit showed 4.02. Despite the high Std. Deviation and lower Mean result, the analysis is still sufficient to conclude that the Mean is 4 and that indicated that the respondents at least agreed with the statement.

For the second statement, we can see some minor improvement on the Mean and the Mean now is 3.89, for *Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan*. The Std. Deviation is .737, -0.092 lower compared to the first statement Std. Deviation. Moreover, the Lower and Upper Limit has a better reading. The lower limit is 3.74 and Upper Limit is 4.04. With all these evidence from the analysis, it can be ascertained that the Mean is 4 and the respondents agreed with the statement. This result is valid enough to justify there is a requirement to combine performance management (RMP Target) and Visual Management (Actual Performance and Trending) to assist both offshore folks and Onshore engineer to ensure the

secondary recovery settings always meet the RMP target and as a result, the production performance is always at an optimum state.

For the third statement, the Mean is 4.08, for *Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)*, +0.19 higher than the second statement and the Std. Deviation is .616. This is an improvement compared to the second statement results despite the Lower Limit still below 4. The Lower Limit results show 3.95 whereas Upper Limit is 4.21. The aforementioned evidence show the Mean is 4 and the respondents agreed with the statement and the conclusion supports Hypothesis 2: Performance Management hybrid with Visual Management will improve performance.

For the fourth statement, the Mean is 4.24 for *Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals*, +0.17 higher than the third statement. The Std. Deviation for this statement is .676. Despite the fact that it showed no improvement on the Std. Deviation, there clearly is improvement on the Lower and Upper Limit results. When comparing Std. Deviation and Lower/Upper Limit results for this statement, it can be assumed a very minor number of respondent answered strongly disagree or disagree. Ultimately, since the Lower and Upper Limit is above 4.0 hence the One Sample *t* Test analysis concluded that Mean is more than 4, thus the respondents agreed with the statement. This evidence is concrete enough to justify that people interpret graphics or visuals better than numbers or texts.

Last but not least, for statement five, the Mean is 4.37 for *Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals* which is the highest among all the statements for hypothesis 2, +0.52 improvements compared to the lowest Mean. It is also visible that the lowest Std. Deviation, .626 among all the statements and it was -.203 reductions as compared to the highest Std. Deviation in hypothesis 2. As a result of higher Mean and lower Std. Deviation, both Lower

Limit and Upper Limit results are more than 4.1. The results verify that Mean is more than 4 and it indicates respondents absolutely agreed with the statement and this conclusion supports the hypothesis 2: Performance Management hybrid with Visual Management will improve performance.

Table 4.9: One Sample T-Test Result for Hypothesis 2

	Null Hypothesis	N	Mean	Std. Deviation	Std. Error Mean	95% CI Lower Limit	95% CI Upper Limit	Conclusion mean is
H2	Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance (to test what is the impact having a tools without the combination of performance management is visual management or vice versa)	91	3.85	.829	.087	3.68	4.02	Mean is 4, indicating the respondent at least agreed with the statement
H2	Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)	91	3.89	.737	.077	3.74	4.04	Mean is 4, indicating the respondent at least agreed with the statement
H2	Having a daily scorecard for platform help to manage performance (i.e. Production, Gas Lift Injection, Water Injection)	91	4.08	.636	.067	3.95	4.21	Mean is 4, indicating the respondent at least agreed with the statement
H2	Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals	91	4.25	.676	.071	4.11	4.39	Mean is more than 4, indicating the respondent agreed with the statement
H2	Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals	91	4.37	.626	.066	4.24	4.50	Mean is more than 4, indicating the respondent agreed with the statement

4.5 Qualitative Findings

Qualitative data concerning how the hybrid or combination of visual and performance management can help to improve organisation's performance were collected through semi-structured interviews with the Asset Manager (member of the senior management team), Operations Manager, Marine Manager, Offshore Installation Manager and Offshore Personnel. The interviews were focused on managerial or management level mainly due to their exposure and knowledge on performance and visual management and this is reinforced with two interviews with the working level. Often, we can see the improvement processes return to their baseline levels after 6 months of implementation despite witnessing improvement trends for the first few months. Sustainability of any continuous performance improvement initiative always remains a challenge for any organisation. Hence, to ensure sustainability on any continuous performance improvement initiative, it had to be driven in a top down manner – meaning, management will have to drive the initiatives. Moreover, the qualitative research will be able to provide a specific answer on the What, How, and Why the hybrid of performance and visual management will be able to help to improve organisation's performance.

Interviews with Asset Manager, Operations Manager, Marine Manager, Offshore Installation Manager and Offshore Personnel did confirm the quantitative result and lend support to the benefits, and the benefits to hybrid both visual and performance management on processes and tools. Partly, these individuals were selected to ensure the comparability of the mixed of operations related personnel and was balanced across the organisation. Table 4.6 lists the three elements and provides supporting quotes from the interviews.

First, the interview revealed a genuine belief from the interviewees that there is a need to have the combination of performance and visual management in managing performance. According to senior management and middle management, performance is all about the basics of having clear goals to achieve the target that has been set. According to the Operations Manager who leading a team of approximately 100 employees to run the platform, outstanding people are

also important to execute the jobs and a clear conversation is crucial to drive the team. Moreover, all the interviews with the senior or middle management provide the same feedback and they agreed that the scorecard with the combination of performance and visual management is able to help them in having a structured conversation about performance with the team. This is mainly due to the scorecard itself being extremely focused on the outcome that really matters to the business. According to Karami et al. (2013), as a result of the granularity of the KPI data displayed on the scorecard, it helps to provide support for the analysis, evaluation, and possible action associated with the potential decision to the team to close the gap. According to one of the middle managers (Marine Manager), the tool also helps them to manage and evaluate contractor performance. Besides, it also helps him and his team to have a transparent conversation with contractors on their performance. For the offshore team, they agreed that the tool and processes help them in having a better performance management of the facility they operate in by providing them a simplified visual of a 7-day trending performance on the platform's critical metrics. According to one of the Offshore Personnel, they now feel engaged and have more sense of ownership when they have visibility pertaining to the organisation's target and goals. They began to prioritize and focus on what is important by looking at the critical metrics on the scorecard. They also began to see and understand what is happening with the scorecards displaying their actual performance against their target.

Finally, the interviews helped explain what performance for them is and more importantly, why and how the combination of performance and visual management can make a difference. Furthermore, the explanations obtained justify how this tool can help them in improving performance and provide them the granularity of performance and transparency within the organisation. All in all, the tools implemented in the organisation support the hypothesis of this study and it is proven that the tools do help in improving communication between team members, help the team stay focused on their goals, overcome impediments, and ultimately improve their operational performance by gaining accountability and ownership of the individual.

Table 4.10: Conclusion and examples from research interviews

Question	Feedback
Asset Manager What Why How	Performance in the end is all about the basics of clear goals, a clear view on the actual and reforecast against those goals. Don't let the system get too "corporate" or complicated because at the end it is about our actual performance against our plans (goals). (What) A clear conversation needs to happen on how to close the gap and drive the team to higher performance. By using a scorecard, I can have a structured and transparent conversation with my managers. (Why) Having a scorecard (the combination between performance and visual management) helps the organisation to ensure we focus on the top 3 or 4 and good things will really happen since human can't focus on 30 things at the same time. (How)
Operations Manager What Why How	Top quartile performance needs clear goals, but most importantly outstanding people who are willing to pull together as a team and not for themselves. (What) It is very important to focus on the outcome that really matters for the business. (Why) A scorecard with the combination of performance and visual management obviously a great tool to drive the right conversation on performance with my team. I can see after the scorecard implementation, the morning call between onshore and offshore had reduced from 40 minutes to 25 minutes. (How) The scorecard itself is extremely focused on the outcome that really matters to the business and because of that, it helps my team to focus on key drivers that enable immediate actions to optimize production. If the scorecard has too much data, then I think it starts to defeat its purpose. (How)
Marine Manager What Why How	Marine strategy and model of operation is slightly different compared to Operations department since all our fleets or vessels are managed by ship owner or contractor. Our performance is actually depending on the contractor performance running the vessel or fleet in supporting the Operations folks. (What)

	For me and my team, we need a process or tools that help us to manage and evaluate our contractor performance. We can see an immediate cascading effect to Operations department when there is any downtime with the fleet or vessel. (Why) Yes, with the scorecard now, it is easier for me to manage those three Floating Storage and Offloading facilities that cost over millions of leasing cost per day and other supply vessel at the location. I can now see three reports in one piece of paper focusing on important metrics and major equipment uptime. This also helps me to have a more structured and transparent conversation with the FSO Master on-board and helps us to align our goals with our contractor. (How)
Offshore Installation Manager (OIM) What Why and How	My performance is measured by the uptime of my facility. Hence, higher uptime and stable production is my goal. I strongly agree that after the implementation of daily scorecard for platform, it does impact how I manage the plant performance. (What) Overall, I think the scorecard is a vital and simplified tool for me to have a better performance management of this facility since the tool now provide me a 7 day trending visual performance of the critical metrics of the platform. It contains all information of the plant performance and issue/concern in hand. (Why & How)
Offshore Personnel Why and How	Before this, I had no idea at all what is our production target and how much production we need to deliver every month. With the existence of scorecard, I finally know the production target for the platform and I began to feel more engaged as a team to ensure high uptime of the plant. I can now work closer with the team to meet the production target because we speak the same language now. I also began to prioritize and focus on what is important based on the scorecard and now I can see and understand what is happening with the scorecard and began take safety as one of my upmost priority. (What & How)

CHAPTER FIVE: DISCUSSION AND CONCLUSION

5.1 Introduction

This chapter will present the discussion on the research findings and analysis written in Chapter 4. Theoretical and practical implications of this research also will be further discussed in this chapter based on the findings. The discussion will include the interpretation of the quantitative and qualitative results whether the hypotheses are supported by the data. Proposal for future research will be outlined at the end of this chapter to overcome the limitations or weaknesses of this research study. Last but not least, this chapter will conclude this research with recommendations for future improvement and lastly summarize if the research has answered the objectives it had sets out to accomplish.

5.2 Discussion

In this research, the researcher wishes to investigate any correlation between performance and visual management and also whether it is a necessity to integrate both of the frameworks in order to achieve the optimum performance improvement results.

The research questions the researcher wished to answer are as follows:

1. Does visual management help to improve organisation performance?
2. Will the combination of performance management and visual management further improve organisation performance?

The results from Crosstabs, One-Sample *t* Test and Qualitative Research have shown significant positive relationship on the ability of performance and visual management in improving organisation's performance. Moreover, this research also shown and proven that with the hybrid of performance and visual management it will further enhance the capability of the tools to improve

performance. Since there was not much of research that had been conducted particularly in relation to the combination of these two popular frameworks, this research will be able to extend the comprehension on the literature of performance and visual management.

Based on the outcomes of the analysis reported, the researcher can confirm that the combination of performance and visual management is capable to further improve the organisation's current performance based on the relatively high overall Mean values from employee perception. Moreover, the perception had been measured with the influence of department, gender and position to ensure there were no biasness and ensure high reliability of the results. According to the findings of this research, all the factors discussed seem to pose significant influence towards the respondents' perception. For instance, the exposure or involvement of respondents on business unit performance management processes greatly impacted their views and feedback. As a result, for the respondents with less or no involvement on the organisation's performance management process, this will greatly hinder the respondent from seeing a bigger picture and experience the combination of performance and visual management in action.

As summarized in Section 4.4, the data analysis of One-Sample *t* Test revealed the overall average Mean score for hypothesis 1 is 4.02 and hypothesis 2 is 4.09. These mean scores showed the general support and agreement by the respondents that Visual Management will improve the organisation's performance. Since the Mean score for hypothesis 2 is slightly higher than hypothesis 1 by +0.07, making it sufficient to support hypothesis 2 that Performance Management hybrid with Visual Management will further improve performance. As noted, education on performance and visual management is crucial and can greatly impacted the employees' perception and the sustainability of the continuous performance improvement initiative. Moreover, the most challenging part is getting used to a new way of working and a new mind-set. This was supported by the crosstabs analysis that showed some of the engineers had difficulty in identifying and realizing the capability and benefits of these tools in assisting the organisation to improve their performance. Later, it was further confirmed by a higher Std. Deviation results from the One-Sample *t* Test analysis

that statement that required adequate understanding from the respondents concerning the process. The researcher strongly believes this gap was also partly contributed by the exposure these respondents received about these frameworks. Many people do not realise the scorecard with the combination of performance and visual management not only provide you information or alert on critical metrics, but also assist the leader in obtaining a structured conversation with his or her team to identify and generate the next course of action to address the issue or identify the root cause and improvement opportunities. Hence, a complete understanding of these frameworks and tools are essential to realise the benefits.

According to the qualitative analysis described in Section 4.5, it provides an idea to the reader on how the combination will be able to improve organisation's performance by focusing on the 'How', 'What' and 'Why'. The interview results were broadly consistent with the earlier research by Bititci et al., (2016) on visual performance management systems that impact organisation practices in which he and the team described.

“Visual strategy and Performance Management approaches are effective in supporting ongoing strategy development and implementation; facilitating performance measurement and review; improving internal and external communication; enabling people engagement in the strategic thinking process; enhancing collaboration and integration among different units and levels; supporting cultural changes and fostering innovation.”

Part of the interview result revealed that to ensure an organisation continues to excel in their performance and efficiency, is important that they only focus on top 3 or 4 high priority area. People tend to lose their focus if they have to focus on too many things as shown in the scorecard. Therefore, the scorecard itself has to focus on metrics that are extremely important for the business that will enable immediate actions to optimize organisation's performance - be it operational or financial performance. If the scorecard has too much data, then it starts to defeat its purpose. That is why it is very important for an organisation to focus on the outcome that really matters for the business. At the end of the day, the top quartile performance needs clear goals, but most importantly clear conversations about how to close the gap and drive the team to higher performance *“Thinking together-Acting together”*

5.3 Recommendations

For practitioners, this research provides value especially for oil and gas industries that are facing uncertain commodity price crisis. From the analysis above, it can be derived that this study to some extent, can positively contribute to an organisation's performance towards the preferred results, corresponding to the discussion as outlined in Chapter 2 – Literature Review of this dissertation. However, there are a few minor weaknesses need to be addressed to make sure it delivers the expected outcomes.

This research is beneficial for any organisation to improve its efficiency and employee engagement while driving lower operations cost baseline and in return, to sustain an organisation's positive free cash flow. As a result, they will also see positive employee engagement and ownership, indirectly reducing turnover rate, increasing productivity and improve organisational performance. Moreover, this research will be able to progressively close the gap between theoretical and practical aspects of performance and visual management. Nevertheless, this research is an attempt to bridge that gap, particularly in the context and setting of an oil and gas firm.

In order to achieve the focus and simplification mentioned in the qualitative research, Key Performance Indicators (KPI) research or study needs to come into the picture. Further studied are required in improving the application skills in working with KPIs to develop scorecards by understanding the concept and application of KPI, the way to produce the right KPI selection with the given organisation's context and KPI selection in a given setting. With that, it gives the organisation a real-world methodology to explore the critical success factors and the groundwork of successful key performance indicators. It is crucial to ensure an organisation is making the right KPI selection and balancing that will allow a precise KPI analysis and focus on the KPI that really matters for the business. Additionally, all these KPIs will be featured in the scorecard. A scorecard's purpose is almost like a car's dashboard. The objective is to provide business unit leader with clear and concise information of key business performance drivers. It helps to provide insight and alert management in a timely manner. With

Performance Management, Visual Management (Scorecard) and selected KPI's, it provides support to the manager in making quick decisions that enables immediate actions to optimize organisation's performance from the high-level overview information of the business. Nevertheless, by having the ability to display these KPI visually and using it to communicate with the workforce, it will ingrain the message into their thinking to gain the accountability and ownership of everyone that is involved in the process.

Last but not least, communication plays a vital role in the sustainability and success of the performance improvement initiative. Management should clarify the purpose of the programme to the workforce and provide more efforts in recognizing employee participation by giving rewards and recognition. Despite the ability of the tools to share strategic information to the organisation, communication is vital to ensure a “common understanding in the company”, as supported by Armstrong (2009) and not to forget, an organisation needs to leverage employee engagement for strategic success. “People” are the key drivers for continual improvement culture in any organisation. Hence, we need to ensure our employees strive for a common objective to improve the company performance and as a result, the impacts will quickly deliver the intended value. Likewise, poor sustainability notwithstanding the initial success can be addressed by improving management and employee engagement, training of supervisors or leaders and members to accomplish a long-term success rate of the program. As a result, this will not only improve organisation performance and ensuring sustainability, but more importantly, improve communication with the team or contractors.

5.3.1 Recommendations for Organisation of This Case Study

When designing a scorecard, the scorecard needs to fit in one page in order to provide an overview with clear and concise information of the key drivers of business performance. Furthermore, the scorecard needs to feature key trends, comparisons against the target and data graphically. By using visual to represent the KPI on the scorecard, it helps to knock out information monopolies and

provoke employees to look at the truth of their own performance (Galsworth, 2005; Liff and Posey, 2004). As a result, it clearly shows the individual's performance whether it is improving, or declining. Finally, by doing this, it helps to provide a precise conclusion that is relevant to the goals of the readers, hereafter the reader them self do not have to waste time to do their own analysis.

Next, to address the low Mean value for one of the statement in hypothesis 1 - Having a daily scorecard for platform or FSO helps to improve the plant uptime, the researcher recommends the company to monitor on the Overdue Safety Critical Element items, Overdue Preventative Maintenance or Overdue Corrective Maintenance. This can be done by designing and implementing a separate scorecard to monitor those metrics since closing out all the SCE, PM and CM items involve a lot of stakeholders. Furthermore, the process itself is complicated since it involves service contract and the availability of spare parts for job execution. Since the process itself involves various stakeholders and for those who are involved in the process, they should understand that Operations department are the customers since they are the ones generating money for the company. According to Tezel et al. (2009), certain metrics related to customer satisfaction metrics need to be publicly displayed. Through visual means, the shared ownership and connection between employees from different departments can be improved (Cropanzano et al., 1997). Hopefully, with the scorecard in place, it will help to create effective process controls that identify problems especially response speed, identify bottleneck, and area of improvement.

Last but not least, based on of the interview with the Offshore Installation Manager, the gas processing plant should have their own scorecard since their operating parameters are too dynamic and processes are too complex to ensure gas production is within the industry specification. According to Ohno, (1988), he also supported that the seven basic visual tools can resolve most of the quality-related issues. Moreover, according to Galsworth, (1997), a scorecard (Visual Aids) is able to translate the complexity to a more easily understood tool by showing the machine status, the plan of the entire operating area with definable objectives and resources. Furthermore, Visual management is a mechanism that enables continuous monitoring, filtering, simplification and distribution of vital

information to individual employees and teams within an organisation. (Suzaki, 1993; Liff & Posey, 2004; Galsworth, 2005).

5.4 Limitations of Study and Directions for Future Research

According to the Malaysian Investment Development Authority (MIDA), it is worth noting that there are over 4000 oil and gas businesses including International Oil Companies, Independents, Service Providers and Equipment manufactures operating locally in Malaysia. This research has focused on one of the selected company out of the 4000 companies. Furthermore, the company had 460 employees with 3 different assets. Out of these, only 92 respondents participated in this survey and only 5 respondents participated in the interview. Thus, the percentage of respondents for this research is relatively insignificant compared to the total number of employees. Since the key here is the people, hence the effectiveness of any performance improvement initiatives lies within its own employee. Subsequent researches also should be carried out in a larger context to improve the effectiveness of the combination of performance and visual management on a wider organisation and not only focusing on the 4 departments – Operations, Subsurface, Drilling and Wells since the ability of the research framework can serve a wide range of functions to the organisation. Moreover, this study does not include the factor of the impacts of wider organisation structure which might affect the design, implementation and outcome of the performance improvement initiative. Therefore, the data collected in the survey and interview might not be reliable enough to represent the entire organisation.

For businesses to meet its goal, it is essential that an organisation ensures an alignment between the three dimensions which are purpose, people and processes and synchronize them with the organisation's strategic and tactical plan. Moreover, this research is a case study and limited to only one company in Malaysia. Hence, different oil and gas company may reveal different outcomes and this framework may not be able to be generalized across all the industries. The applications of this framework can be researched for different companies in the industry that are operating with a different business model. A cross- company

learning effort is required for a broader application and understanding of the application with possible new functions or modifications might be required especially on how other organisational apply this framework.

Though there are many areas within this research topic for forthcoming discussion, a couple of the key areas which can be further investigated are the performance measure and target setting that greatly impact the organisation's behaviours. Not forgetting communication, performance dialogue is very important in order to extract the full benefit of this framework and tool. Hence, it is vital for any organisation to have regular meetings to review the scorecard that is focused on the business – not on individual performance. During the dialogue, pertinent questions need to be asked and the examples are listed below.

- What is happening? (Identify gaps to target)
- Why is it happening? (What happened that cause the performance gaps/Root causes)
- What needs to be done? (Mitigation plan/Root Cause analysis to prevent reoccurrence)
- Who is going to do it? (Responsibility/Ownership)
- When is it going to be done? (Priority/Deadline)

Effective long-term application of performance and visual management system requires a constructive organisational culture. The linkages of the research framework with other management systems component need to be understood and clearly stated to ensure the processes are properly connected to achieve the optimum results.

Finally, confirming the results by Tezel et al. (2009), Visual Management is capable to assist companies in experiencing cultural transformation to be more effective and efficient in the way they work. The framework has proven its effectiveness in promoting cultural change, “building a continuous improvement culture” and enhancing innovation process. The methodology of combining performance and visual management is a powerful process and tool if it is used in

the correct manner. It also helps to change the mind-sets about “Practice the philosophy of continuous improvement. Get a little bit better every single day.”

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APPENDICES

Re: U/SERC/70/2017

11 September 2017

Dr Pok Wei Fong
Department of Economics
Faculty of Accountancy and Management
Universiti Tunku Abdul Rahman
Jalan Sungai Long
Bandar Sungai Long
43000 Kajang
Selangor

Dear Dr Pok,

Ethical Approval For Research Project/Protocol

We refer to your application dated 22 August 2017 for ethical approval for your research project (Master student's project) and are pleased to inform you that your application has been approved under expedited review.

The details of your research project are as follows:

Research Title	Visual Management, Performance Management and Performance Improvement: A Case Study of An Oil and Gas Company in Malaysia
Investigator(s)	Dr Pok Wei Fong Young Wei Loong (UTAR Postgraduate Student)
Research Area	Social Sciences
Research Location	Kuala Lumpur
No of Participants	50 participants
Research Costs	Self-funded
Approval Validity	11 September 2017 - 10 September 2018

The conduct of this research is subject to the following:

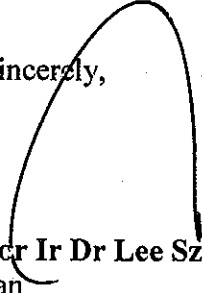
- (1) The participants' informed consent be obtained prior to the commencement of the research;
- (2) Confidentiality of participants' personal data must be maintained; and
- (3) Compliance with procedures set out in related policies of UTAR such as the UTAR Research Ethics and Code of Conduct, Code of Practice for Research Involving Humans and other related policies/guidelines.

Should you collect personal data of participants in your study, please have the participants sign the attached Personal Data Protection Statement for your records.

The University wishes you all the best in your research.

Thank you.

Yours sincerely,



Professor Ir Dr Lee Sze Wei
Chairman
UTAR Scientific and Ethical Review Committee

c.c Dean, Faculty of Accountancy and Management
 Director, Institute of Postgraduate Studies and Research

Section A: Respondent Demographic Profile (please tick which applicable)

Department : ☐ 1.) Operations ☐ 2.) Subsurface ☐ 3.) Drilling ☐ 4.) Wells

Gender : ☐ 1.) Male ☐ 2.) Female

Position : ☐ 1.) Management ☐ 2.) Engineer ☐ 3.) Offshore

Section B: General Question (please CIRCLE one answer for each question)

	Please SELECT the most appropriate number: 1=Strongly Disagree ; 2=Disagree ; 3=Neutral ; 4=Agree ; 5=Strongly Agree	Please CIRCLE your selection				
1	Having a scorecard (visual management) helps to manage performance in general (<i>to test the 'does'</i>)	1	2	3	4	5
2	Having a scorecard (visual management) helps to provide better visibility that is required in managing performance (<i>to test the 'how'</i>)	1	2	3	4	5
3	Having a scorecard (visual management) helps to simplify performance management processes	1	2	3	4	5
4	Having a scorecard (visual management) helps to identify ways to improve by providing the information that is required for improvement (<i>to test the 'how'</i>)	1	2	3	4	5
5	Having a scorecard (visual management) helps to provide organization the clarity on performance in comparison to business unit goals (<i>referring business unit weekly bulletin</i>)	1	2	3	4	5
6	Performance management CANNOT achieve its full potential without a visual management (scorecard - graph/traffic light) that provides visibility to manage performance (<i>to test what is the impact having a tools without the combination of performance management is visual management or vice versa</i>)	1	2	3	4	5

Section C: Operations / Production

	Please SELECT the most appropriate number: 1=Strongly Disagree ; 2=Disagree ; 3=Neutral ; 4=Agree ; 5=Strongly Agree	Please CIRCLE your selection				
1	Having a daily scorecard for platform or FSO helps to improve the plant uptime	1	2	3	4	5
2	Having a daily scorecard for platform helps to meet the requirement or target for Reservoir Management Plan (RMP)	1	2	3	4	5
3	Having a daily scorecard for platform help to manage performance (ie. Production, Gas Lift Injection, Water Injection)	1	2	3	4	5
4	Having a daily scorecard for platform or FSO help to provide better monitoring (trending) and review of progress and results compared to normal daily reports which only indicate NUMBERS – No Visuals (<i>referring daily scorecard vs daily production reports</i>)	1	2	3	4	5
5	Having a daily scorecard helps onshore support teams to provide the necessary support required by offshore crews (<i>to test the 'how'</i>)	1	2	3	4	5
6	Production target (Op-Plan) set on the scorecard is reasonable and achievable	1	2	3	4	5

Legend

	Hypothesis 1
	Hypothesis 2

Weekly Delivery Bulletin - Week of Oct 26 - Nov 1, 2015

Weekly Operational Commentary

HSE:

I am pleased to report another good week, with no incidents or injuries. Well done, and thank you for your continued hard work. We have now entered the final two months of 2015 and we all need to stay focused on our primary goal, which is Safe Delivery, to ensure that we finish the year off on a positive note.

Production:

Overall, production was on target for the week. Peninsular Malaysia was above target due primarily to strong oil production from Platform 1. Gas production was also above target, as a result of higher gas demand from buyers. Platform 2i was below target due to platform ESD. Platform production was also below target because of an unplanned third party shut down

HS&E Incident Reports - As at 11 October 2015

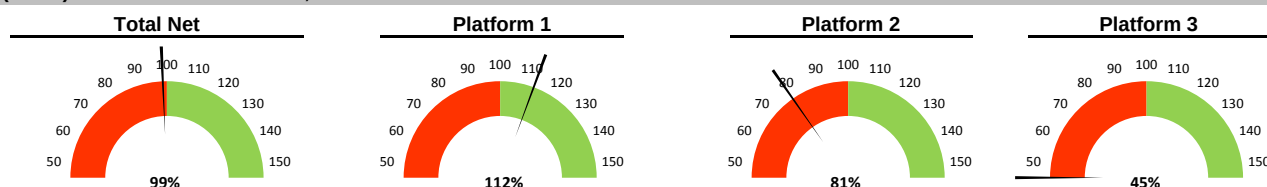
	Days since last	
High Profile Incident	146 ☺	
Lost Time Injuries	55 ☺	
Total Recordable Cases	15 ☺	

	Days since last	
Environmental Spills	624 ☺	
Hydrocarbon Releases	277 ☺	

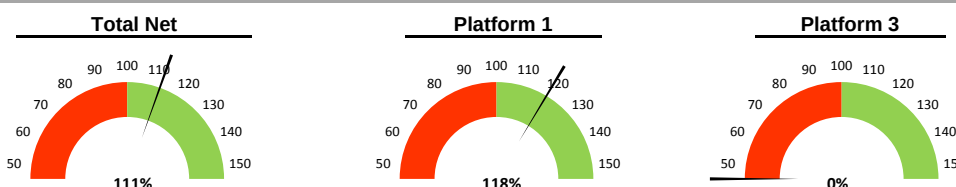


Weekly Net Production Target - % Achieved vs. Plan

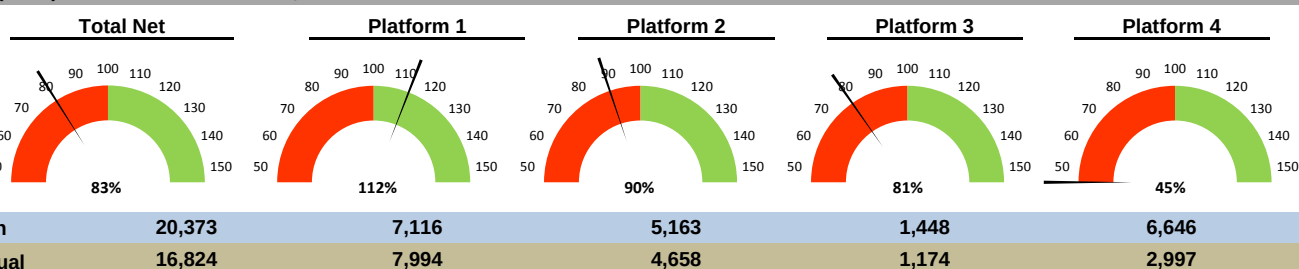
Net (boe/d) - Week of Oct 26 - Nov 1, 2015



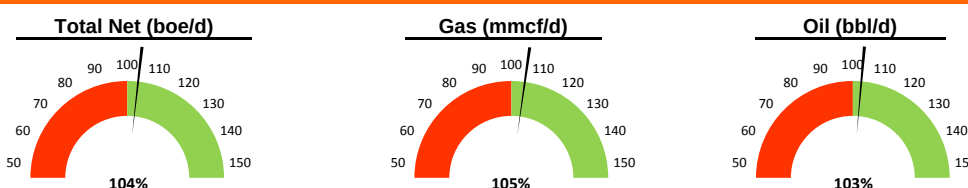
Gas (mmcf/d) - Week of Oct 26 - Nov 1, 2015



Oil (bbl/d) - Week of Oct 26 - Nov 1, 2015



YTD Net Production Performance



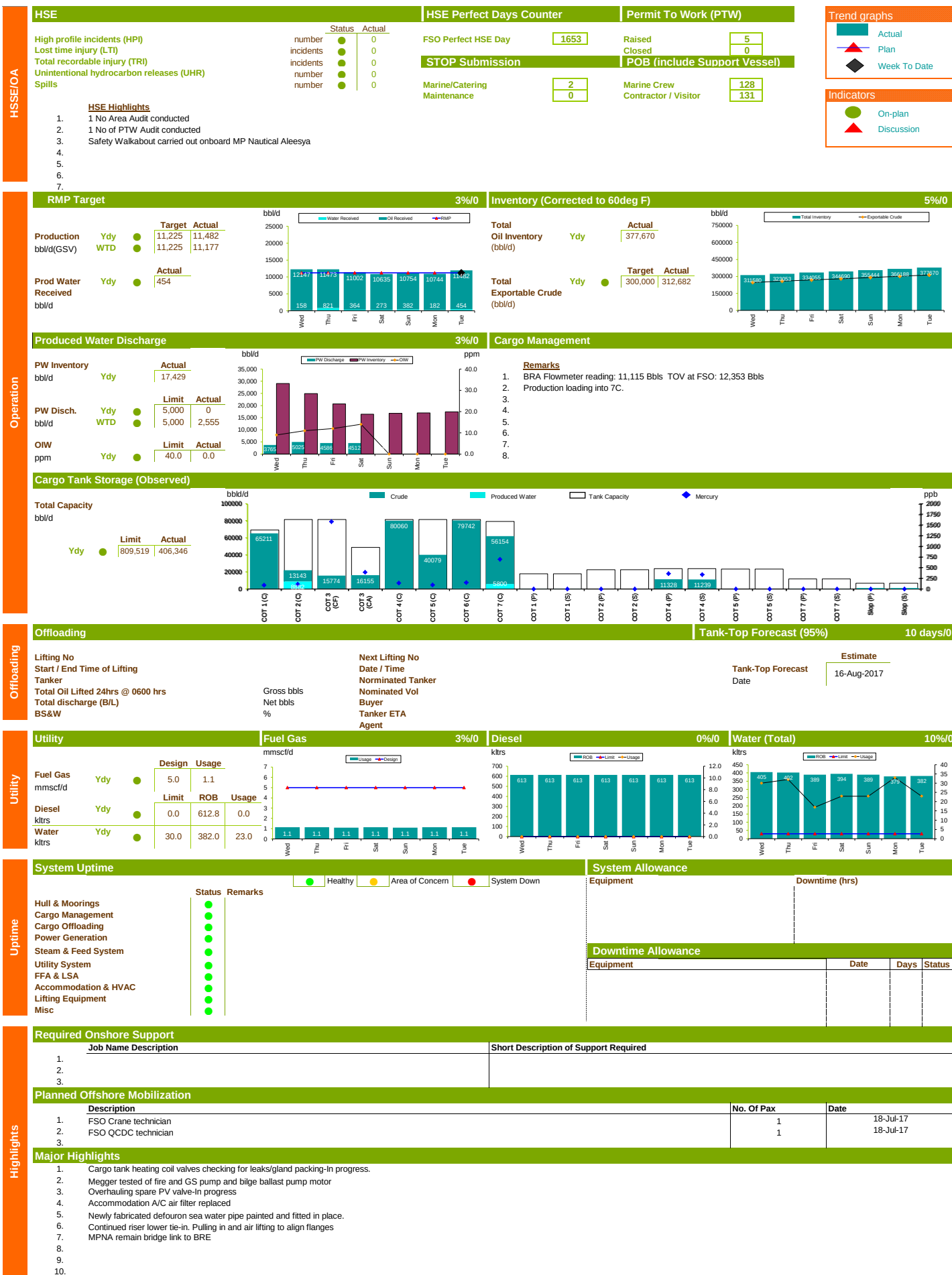
LEGEND

Below Target

On / above Plan

Plan

Actual



Platform 1 Daily Scorecard

ver 1.5

Tuesday (Sep 17, 2013)

HSE OI

HSE

High profile incidents (HPI)
Lost time injury (LTI)
Total recordable injury (TRI)
Unintentional hydrocarbon releases (UHR)

	Status	Actual
number	●	0
incidents	●	0
incidents	●	0
number	●	0

Flare (mmscf/d)
Flare (mmscf/d)
CO2 Vent (mmscf/d)
OIW (ppm)
OIW (ppm)
Spills

	Status	Limit	Actual
	●	25	2
	●	290	173
	▲	25	83
	●	25	1
	●	0	0

Trend graphs



Indicators



HSE Highlights

- STOP cards submission is 5 : BRE Maint- 1, BRE Prod- 3, Const- 1 .
- PM3-South Complex HSE Perfect Day - 36 days.
-

Production

		Target*	Actual		Nom	Actual
Net Liquid	Ydy	16,976	16,781	PN	50.0	54.6
bbl/d	WTD	16,976	16,439	PV	190.0	184.3
Net Sales	Ydy	210.0	239.0			
Gas	mmcsf/d	210.0	235.1			
Gross	Ydy	230.8	263.1			
Gas	mmcsf/d	230.8	258.2			

Gas Nomination

mmcsf/d

3%/0

Condensate

bbl/d

3%/0

CO2 %

%

CO2 %

%

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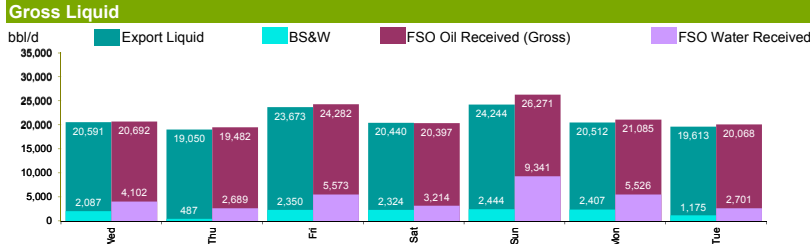
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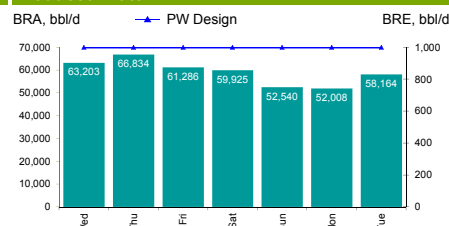
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Gross Liquid



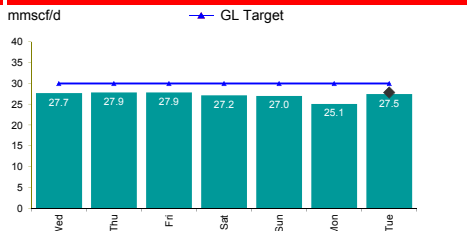
Produced Water



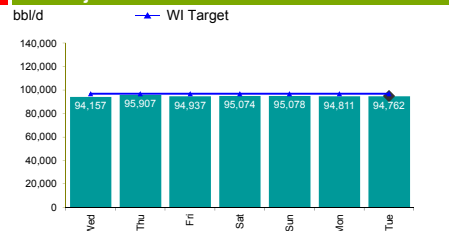
Secondary Recovery

	Dsgn	Target	Actual
GL	20.0	30.0	27.5
mmcsf/c			
WI	122,000	97,000	94,762
bbls			

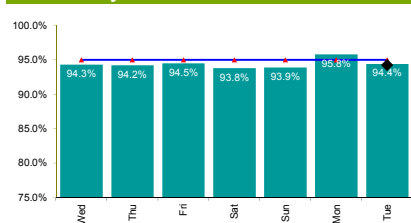
Gas Lift Volume



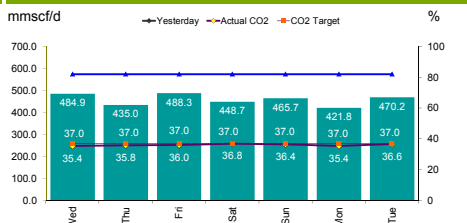
Water Injection Volume



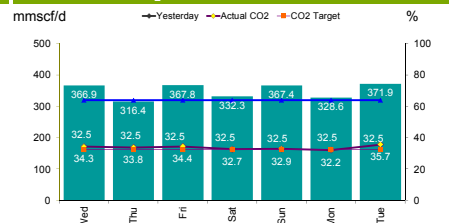
HC Recovery



Feed Gas & CO2%



Gross Gas & CO2%



Unplanned Losses

Description of Losses

- Loss of WI due to unavailability of BRAU WI P-5155B (24 hrs loss).
-
-
-
-
-
-

Est Losses

11557 bbls WI

Production & Optimization

- Hg monitoring - NF Condy: 377 (ppb), NF Gas: 1 (µg/Nm3) ; Crude FSO: 54 / 56 (ppb).
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Required Onshore Supports

Job Name Description

Short Description of Support Required

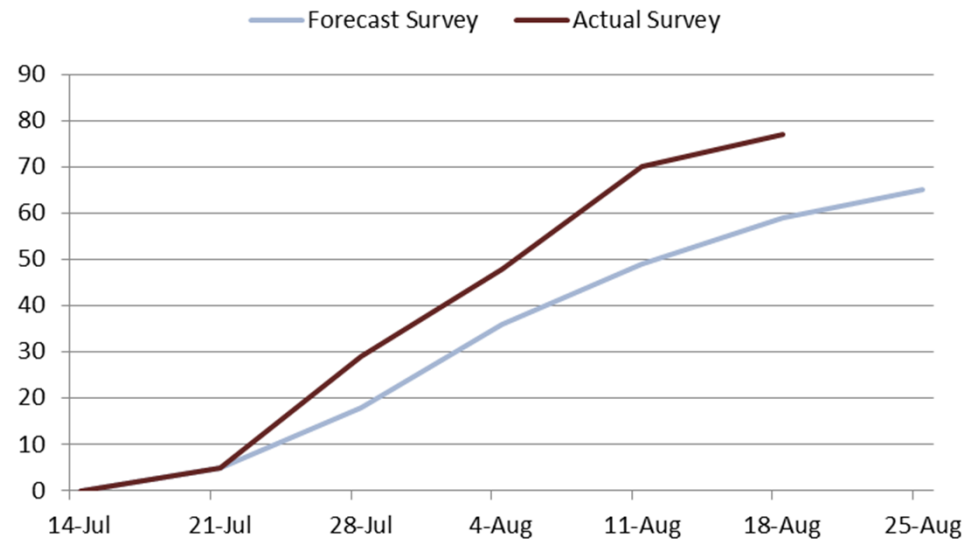
Major Highlights

- FSO lifting / inventory: Exportable dry crude inventory as of 0600 hrs is 208,319 bbls. Next lifting tentative date 28-30th Sep with parcel of 300K bbls.
- OIW reading: Average 24 hrs Lab reading is 83 ppm.
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Survey Progress Weekly Report

Stephen Young MBA Dissertation Survey Progress)

Report Date: 18-Aug-2017



Department	Progress (14 Jul -18 Aug)		Plan	Actual	
Operations	●		30	37	Complete
Offshore	●		15	15	Complete
Subsurface	●		10	17	Complete
Wells	●		5	4	In-Progress
Drilling	●		5	4	In-Progress
Total	●		65	77	In-Progress

Weekly Summary: Survey collection completed for Offshore. Target to complete survey collection for Drilling and Wells end of this week. Forecast total survey collected by end of this week will be 80 samples.